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Data Communication and Computer Networks

NETWORK CHARACTERIZATION

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CONTENTS

Network Characterization:

- Introduction & Data Communication Concepts
- Goals and Applications
- Network Design-Transmission Technologies
- Categorization according to Size, Purpose, Design issues & Transmission Technologies

INTRODUCTION

- **A *Computer Network* may be defined as "*an interconnected collection of autonomous computers*".**
- **Two computers are said to be interconnected if they are able to exchange information.**
- **The connections may be thru copper wires, optical fibers, and wireless electromagnetic or optical media.**
- **'Autonomous' means that there is no master/slave relationship between the connected devices.**



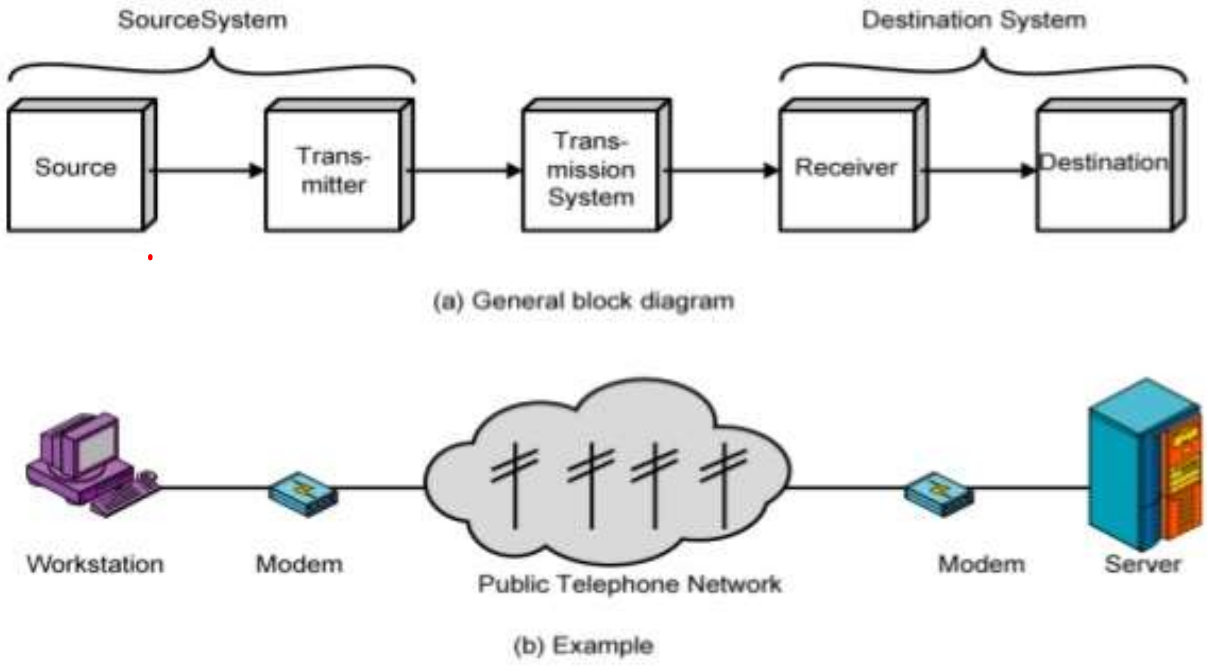
Terminal



Workstation

DATA COMMUNICATION CONCEPTS

- **Data Communication** is the exchange of data (in the form of 0's and 1's) between two devices via some form of transmission medium (such as a cable).
- **A communication system** is a set of devices (hardware and software) whose purpose is sharing data.



Goals and Applications of Computer Networks

For Companies/Organizations

A computer network must satisfy several key criteria for optimal functionality:

- **Fast means of sharing and transfer of files/data or sharing of load**
- **Centralized software management**: streamlines software tasks by administrating, deploying, and maintaining applications from a central point in computer networks, improving control and coordination across devices.
- **Cost effectiveness:** achieving valuable results by efficiently utilizing resources, finances, and operational efforts within the network infrastructure and related tasks.
- **fast access:** Performance, assessed by transit time (message travel duration) and response time (enquiry-to-response delay).
- **Scalability: the** ability to smoothly expand capacity, performance, and services to meet growing demands and users' needs.
- **Reliability:** indicating the likelihood of error-free operations within a defined timeframe.
- **Security:** ensuring the safeguarding of data against unauthorized access.

Goals and Applications of Computer Networks

- **Access to Remote Information:**
  - Empowers people, businesses, and societies with global access (regardless of physical location)
  - Support for online banking, shopping, news, and libraries
  - The World Wide Web provides diverse information, including art, business, and health
  - Online newspapers, journals, and digital libraries are also available

- **Person –to-person Communication:**

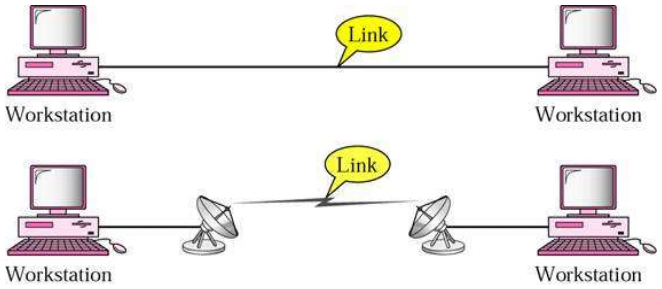
  Achieved through video conferencing, email, messaging etc
  - Enables global connectivity and real-time interaction
  - Webinars/live streams, offering diverse content to global audiences
  - Enhances collaboration and supports social networking
  - Supports distance learning and language acquisition
  - Aids emergency response and global awareness
  - Promotes cultural exchange and personal growth.

- **Interactive Entertainment:**
  - Real-time gaming
  - Social connectivity
  - Diverse game modes, massive multiplayer experiences
  - Innovative platforms
  - Content distribution:
    - streaming services (Netflix, YouTube)
    - podcasts (Apple Podcasts, Spotify)
    - digital books (Amazon Kindle)
    - online news outlets, app stores (Google Play, Apple App Store)
    - social media platforms (Facebook, Instagram)
    - online courses (Coursera, Udemy)
    - cloud storage services (Google Drive, Dropbox)
  - E-sports
  - Continuous updates
  - Cross-cultural exchange

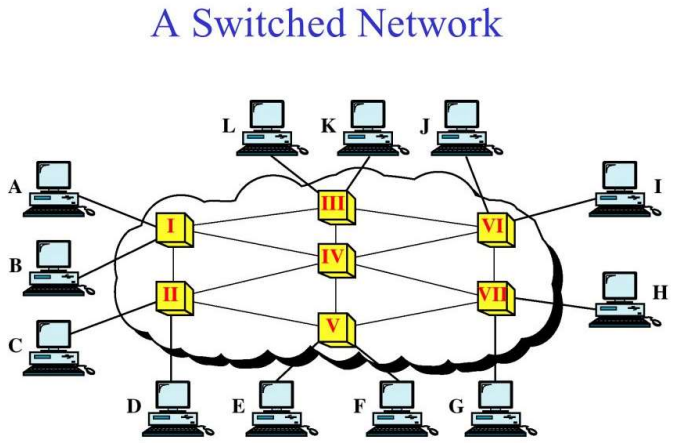
Network Design – Transmission Technologies

Point-to-Point (Store-and Forward / Switched Transmission Technology)

multiple communication channels (mostly in WANs)

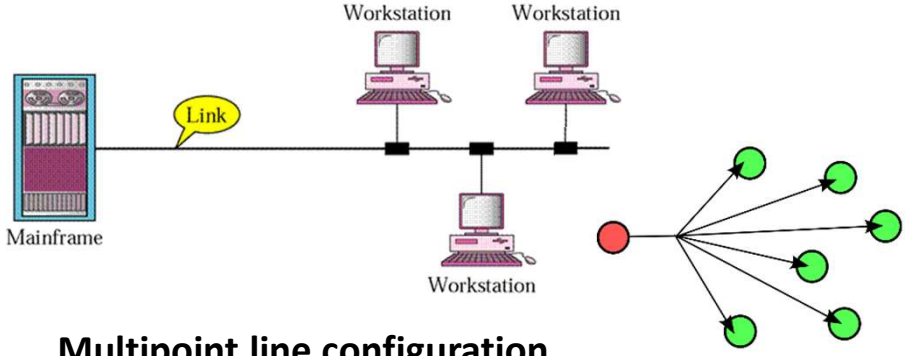
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Point-point line configuration

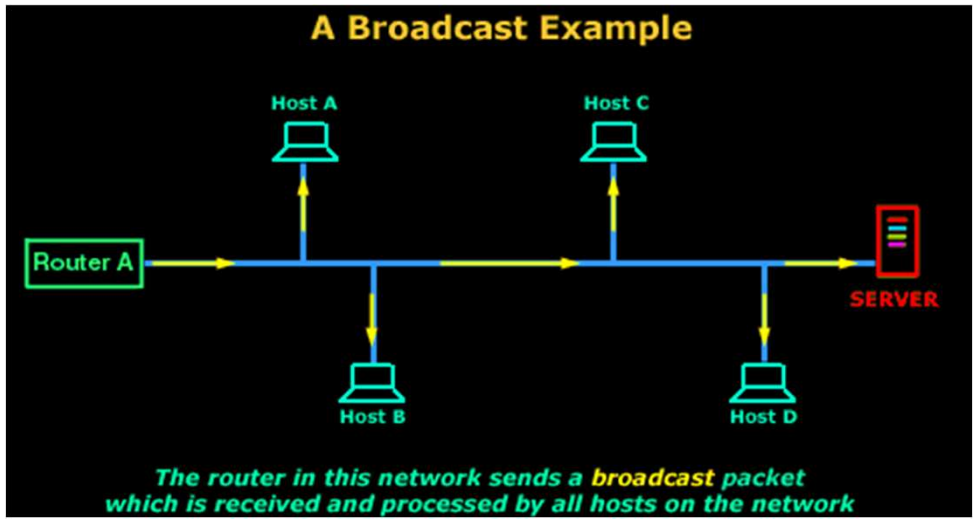


Multipoint (Broadcast Transmission Technology)

single communication channel (mostly in LANs)



Multipoint line configuration



Categorization of Networks

Based on size

- Personal area network (PAN)
- Local area network (LAN)
- Metropolitan area network (MAN)
- Wide area network (WAN)

Based on purpose

- Storage area network (SAN)
- Enterprise private network (EPN)
- Virtual private network(VPN)

Personal Area Network (PAN)

Computer network organized around an individual person in a small office or residence

PANs can be used for communication among the personal devices themselves (intrapersonal communication), or for connecting to a higher level network and the Internet.

PAN may include wireless computer keyboard, mouse, Bluetooth enabled headphones, wireless printers, TV remotes etc

Examples are Bluetooth, Zigbee, Wi-Bree, IRDA, RFID, NFC, Wireless USB, UWB

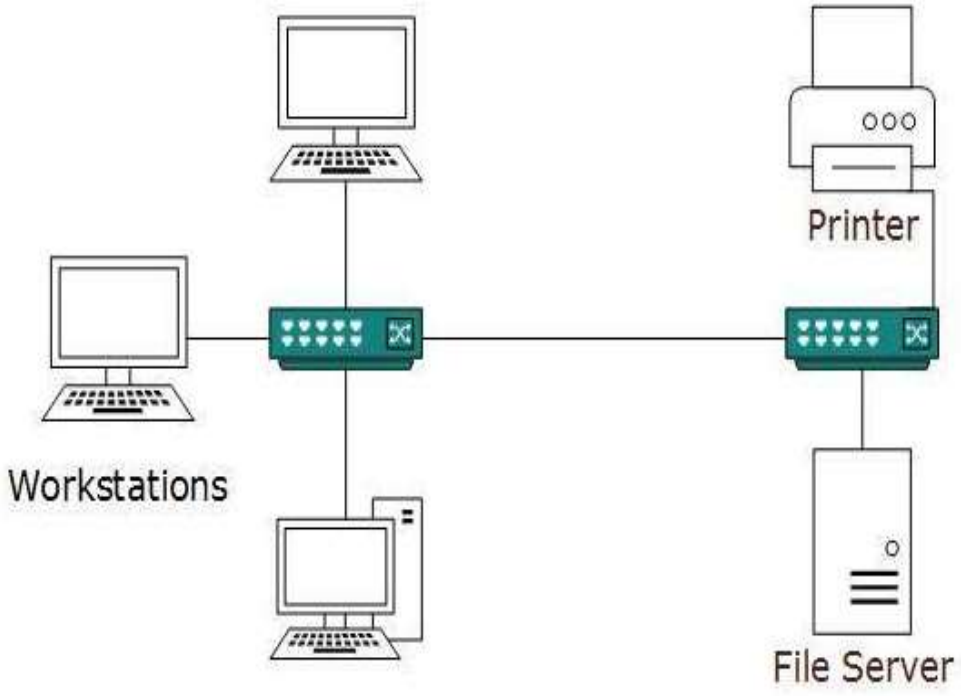


Local Area Network (LAN)

- Devices in a limited geographical area such as home, school, computer laboratory or office building
- Can be Wired or Wireless
- High speed and relatively low cost
- Use only one type of transmission medium
- Transmission technology used is broadcasting
- Design issue - arbitration mechanism (Media Access Control) to resolve conflicts amongst multiple simultaneous transmissions over the single media.

COMPONENTS OF LANs

- **Computers** and other devices such as **printers**
- A relatively powerful computer which functions as the network **"server"**
- A **communications channel** connecting the computers and devices.
- **Network Interface Card** (also referred to as an adapter/LAN card) for a computer or Device to be connected to the LAN.
- One or more **Switches/Hubs**
- Popular LAN operating Systems (**Novell NetWare** and **Windows 2000 server, Linux, Unix)**

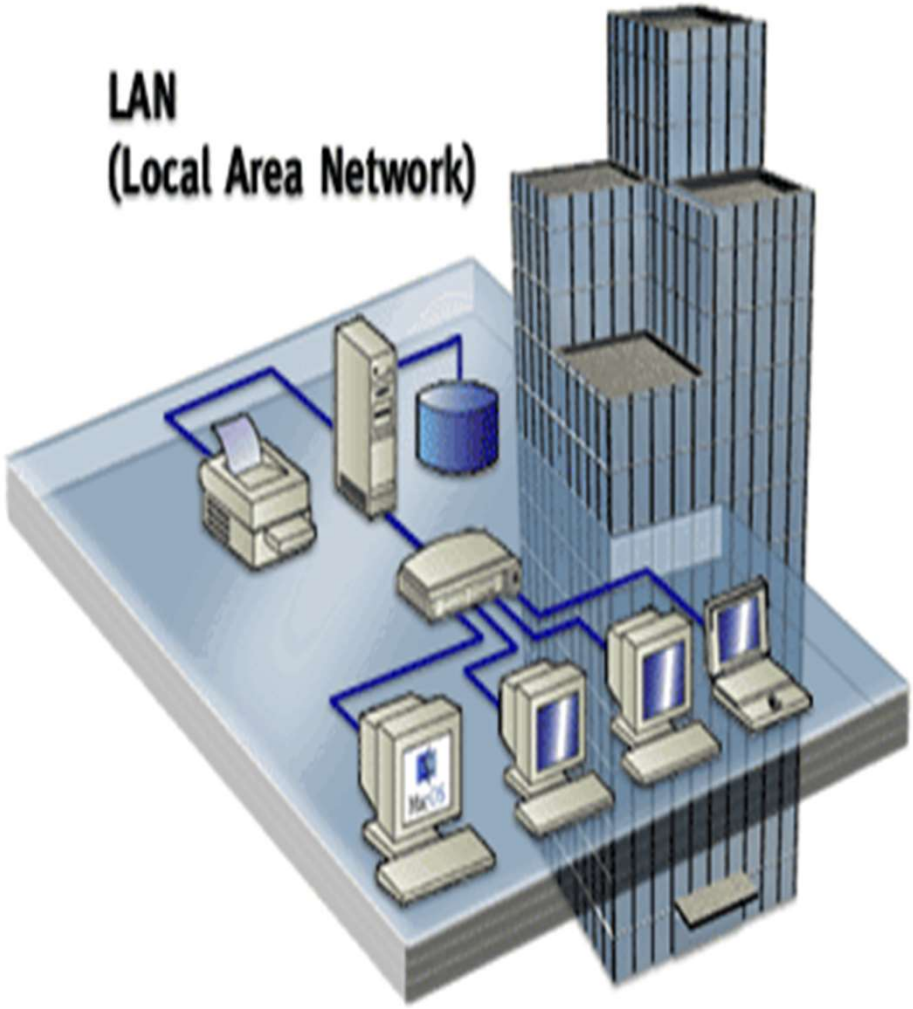


Characteristics of LANs

- **LANs have data rates in the 4 to 16Mbs range. Today, however, speeds are increasing and can reach 100 and 1000 Mbps with gigabit systems**
- **A local area network (LAN) is usually privately owned and links the devices in a single office, building or campus.**
- **Typically used for single sites where people need to share resources among themselves but not with the rest of the outside world**
- **If a LAN is entirely wireless, it is referred to as a wireless local area network, or WLAN.**
- **Common topologies are Bus, Ring, Star**
- Earlier common LAN standards - **ARCNET, Token bus, Token Ring** etc.
- Current technologies - **IEEE 802.3 Ethernet** over twisted pair cabling, and **Wi-Fi (wireless IEEE 802.11)**

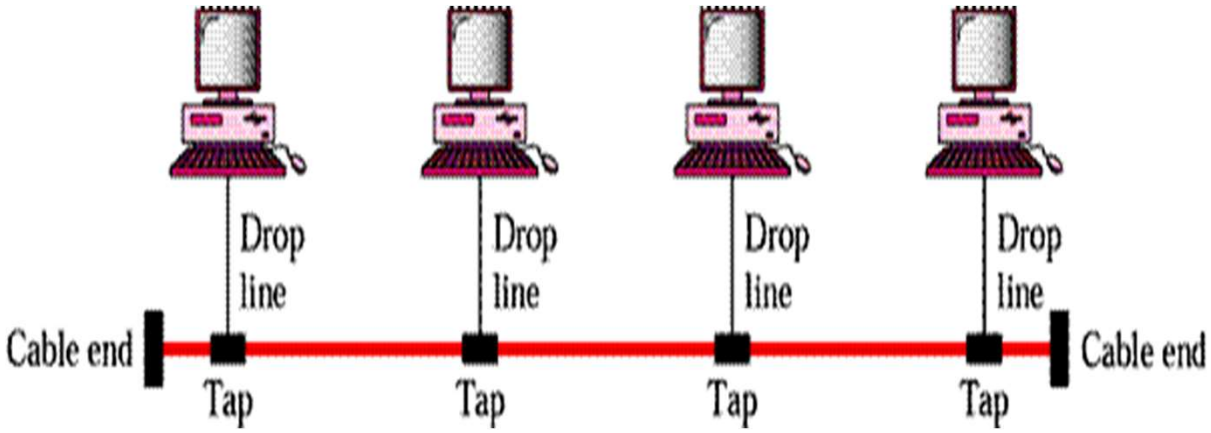
LOCAL AREA NETWORK



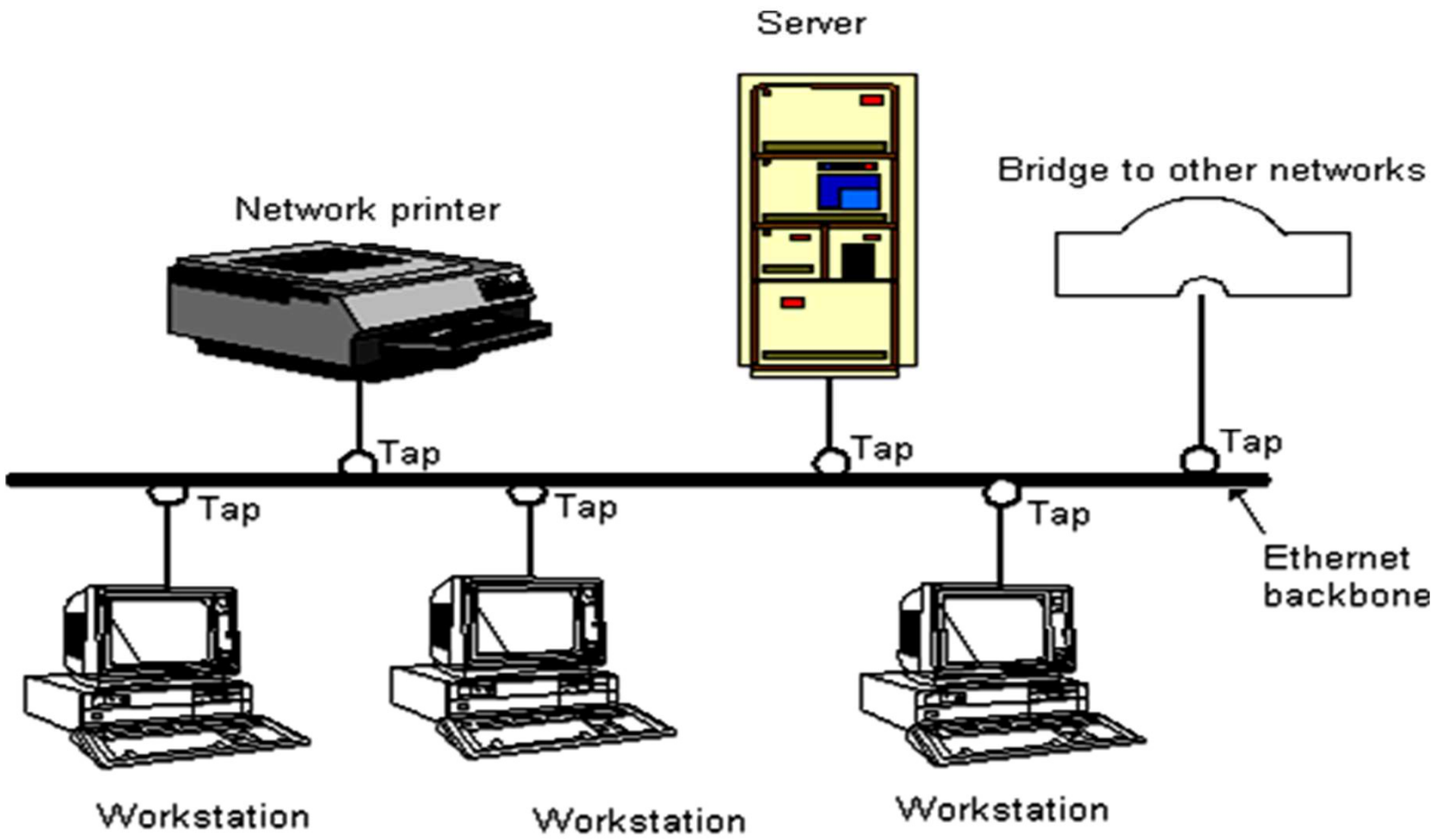


BUS Topology for LANs

- **Bus topology is the most common type of multipoint topology used in local area networks.**
- **Bus topology is characterized by one starting point and one ending point.**
- **Each computer on the network has just one connection. This connection is attached to a single cable that is shared by all other devices on the network.**
- **A bus network topology works well for smaller networks and is inexpensive to install. However, it can become slow when more devices are added to the network.**
- ***Advantages*: easy to install** and **uses less cabling** than mesh, star, or tree topologies. ***Disadvantages:* difficult reconfiguration and fault isolation**. **difficult to add new devices**



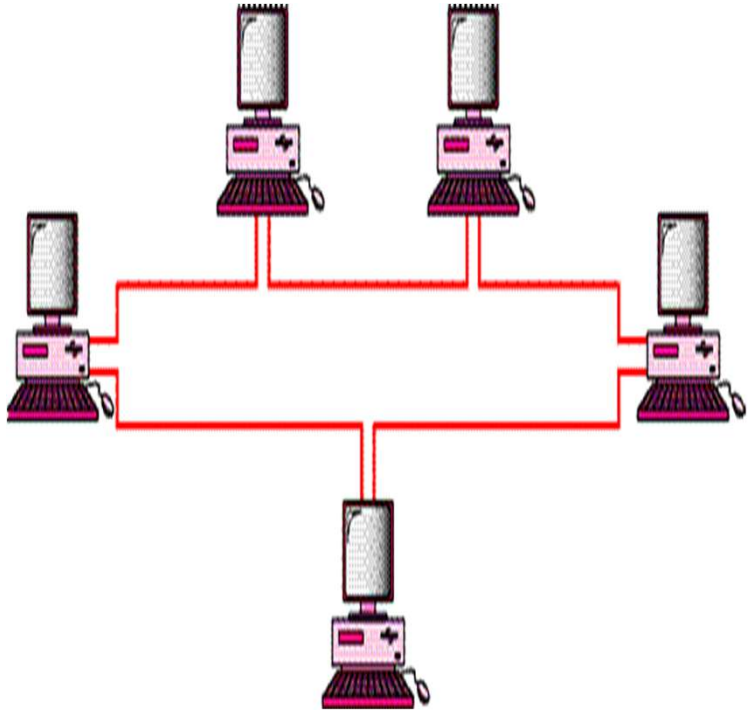
BUS Topology - Egs. IEEE 802.3 Ethernet LAN standard using coaxial cables (10Base2 and 10Base5) , IEEE 802.4 Token Bus



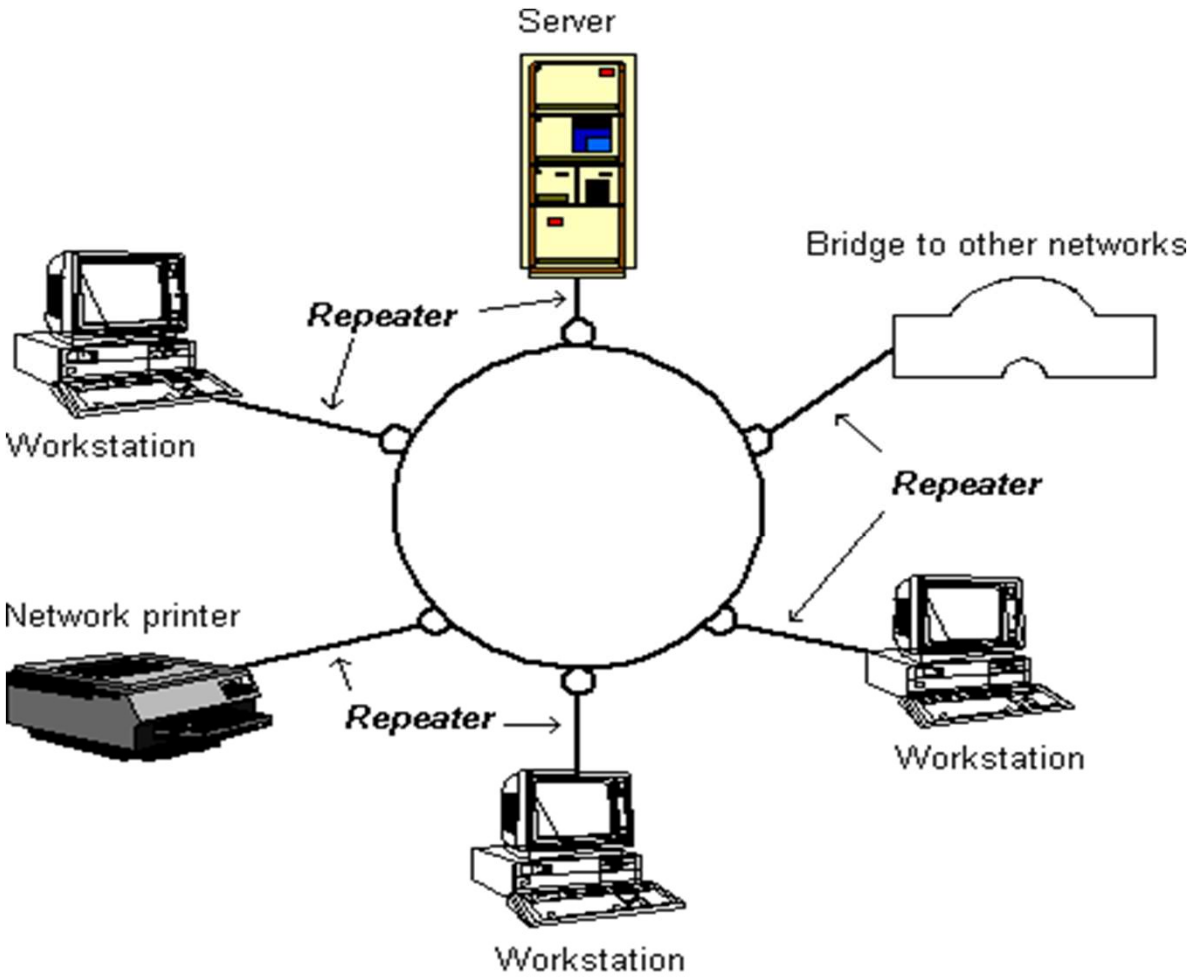
RING Topology for LANs

- Ring is a circle with no ends.
- All the devices on the network are connected to the ring.
- Works well for a computer network with many devices attached to it.
- Each device has a dedicated point-to-point line configuration only with the two devices on either side of it.
- ***Advantages :*** easy to install and reconfigure. To add or delete a device require only two connections. Easy fault isolation.
- ***Disadvantage:*** A break in the ring (such as a disabled station) can disable the entire network.

- Eg. IEEE 802.5 Token ring LAN

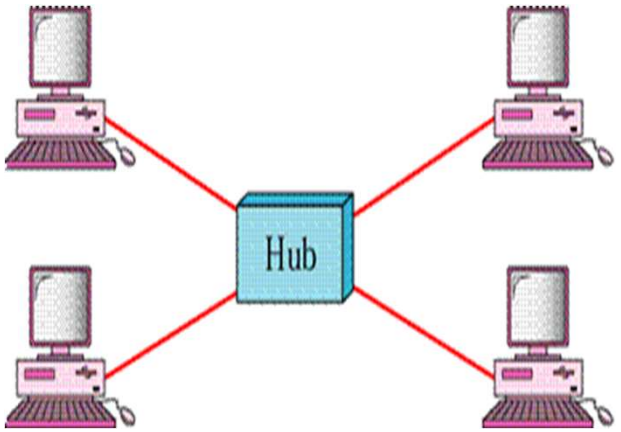


RING - Eg. – IEEE 802.5 Token Ring

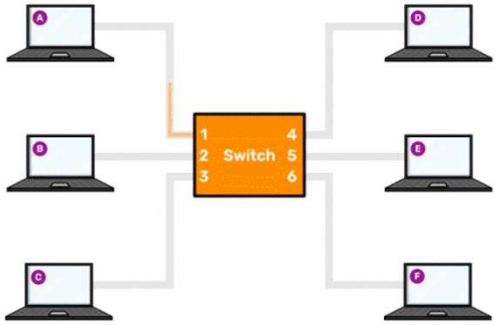


Ring network

STAR TOPOLOGY



- A **star topology** describes a network in which all the devices are connected to a central device, which may be a hub or a switch, through dedicated links.
- **A star topology does not allow direct traffic between devices. If one device wants to send data to another, it sends the data to the central hub or switch, which then relays the data to the other connected device**
- **Example is IEEE Ethernet LAN (10BASE-T, 100BASE-T)**



Advantages:

- ❑ **It is easy to install and reconfigure.**
- ❑ **Troubleshooting and management is the easiest in this topology.**
- ❑ **The star topology is robust. If one link fails, only that link is affected. This factor also lends itself to easy fault identification and fault isolation.**

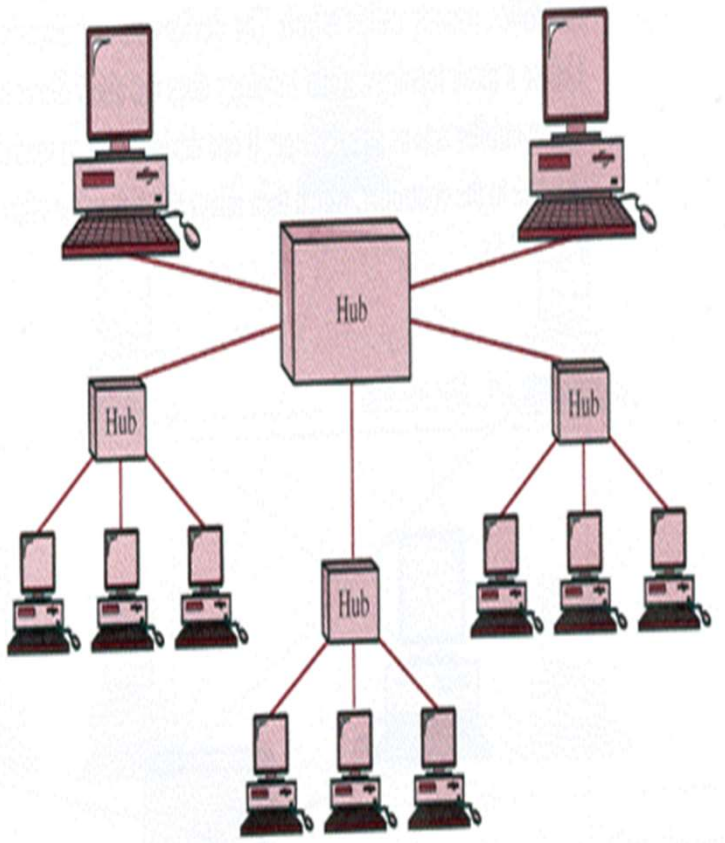
Disadvantages:

- ❑ **A big amount of cabling and a big number of I/O ports are required, i.e, the hardware required can be very expensive but Star topology requires far less cable than Mesh topology.**

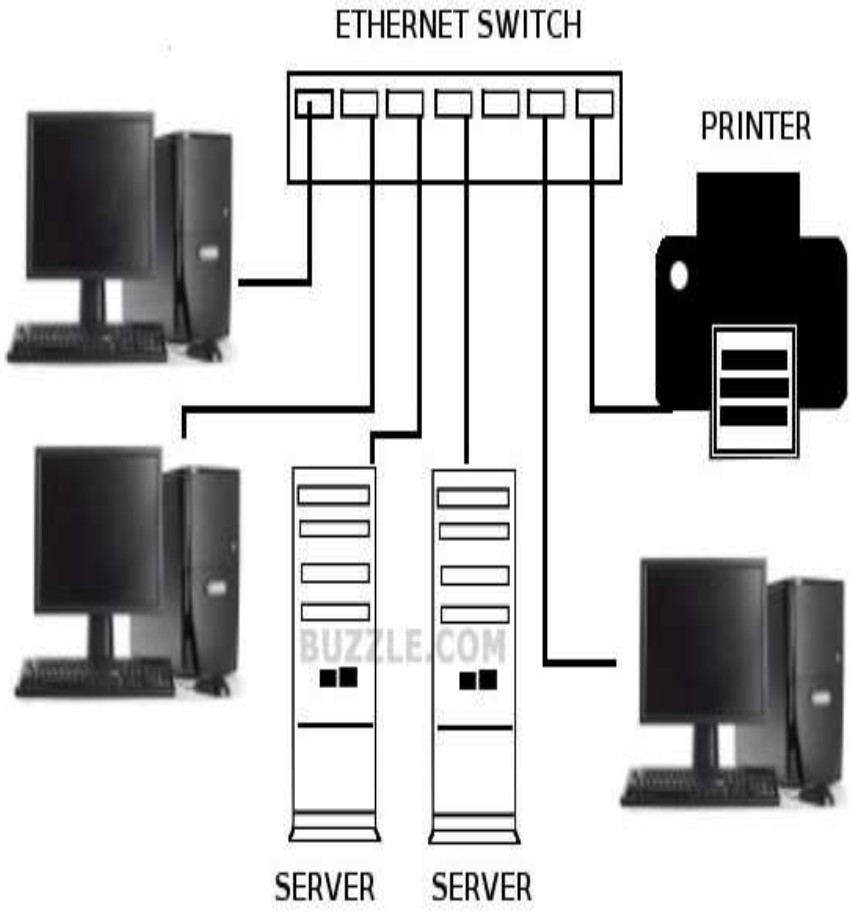
Tree Topology

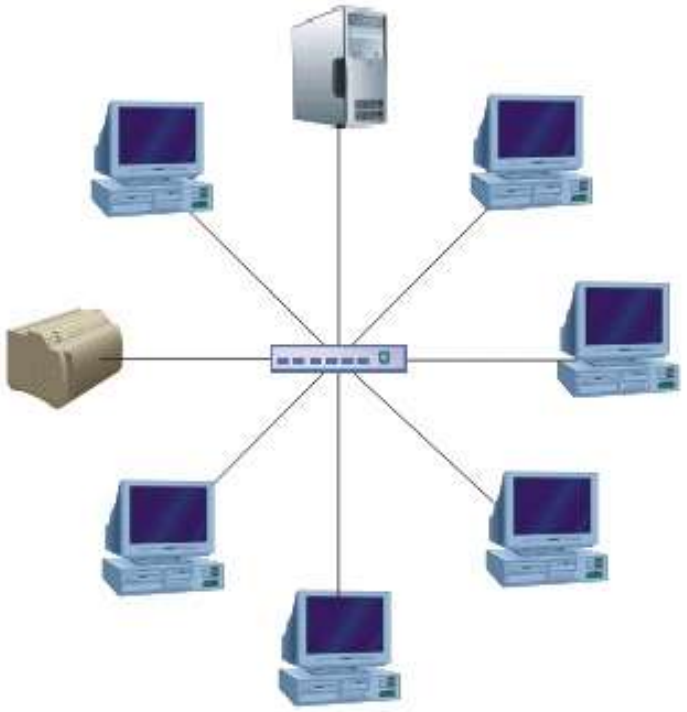
(mostly used in MANs)

- **A tree topology is a variation of a star.**
- **Nodes are linked to a central hub that controls the traffic to the network.**
- **The majority of devices connect to a secondary hub that in turn is connected to a central hub.**
- **The central hub in a tree is an active hub. An active hub contains a repeater, which is a hardware device that regenerates the received bit patterns before sending out. Repeating improves the transmission quality, and increases distance that a signal can travel.**
- ***Avantages and disadvantages* of a tree topology are generally the same as those of a star.**
- **Example: cable TV technology (MAN) where the main cable from the main office is divided into many branches and each branch is divided into smaller branches and so on.**

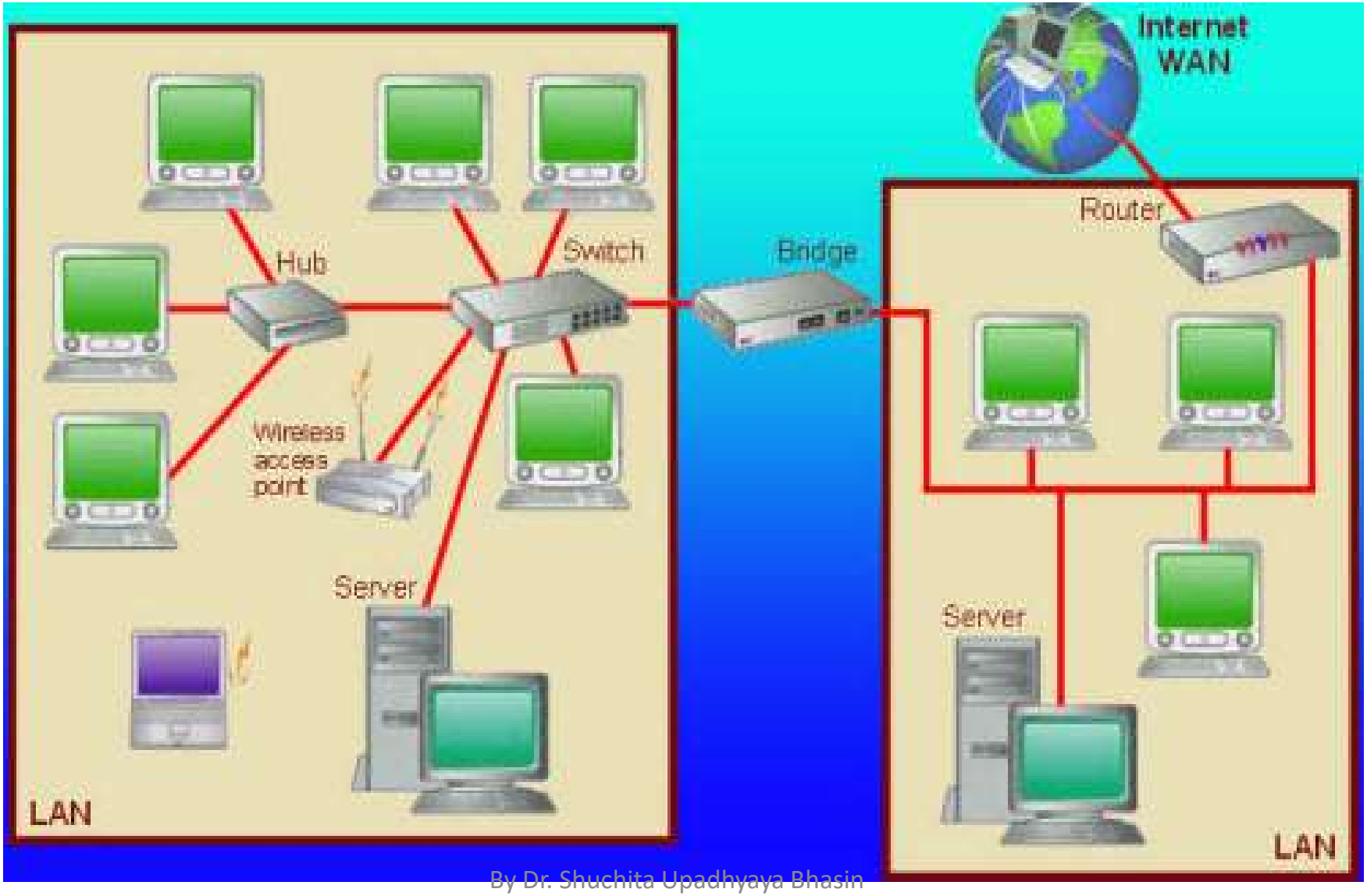


STAR Topology – eg. IEEE 802.3 Ethernet LAN standard (10Base T and 100Base T)





LAN with Internet Access



LAN – HOME AREA NETWORK

If multiple individuals use the same network within a residence, the network is sometimes referred to as a home area network, or HAN

Connects

- Computers
- Peripheral devices
- Telephones
- Digital recorders
- CCTV & televisions
- Video games
- Home Security Systems
- "smart" appliances,
- Fax machines

In a very typical setup, a residence will have a single wired Internet connection connected to a modem. This modem then provides both wired and wireless connections for multiple devices.

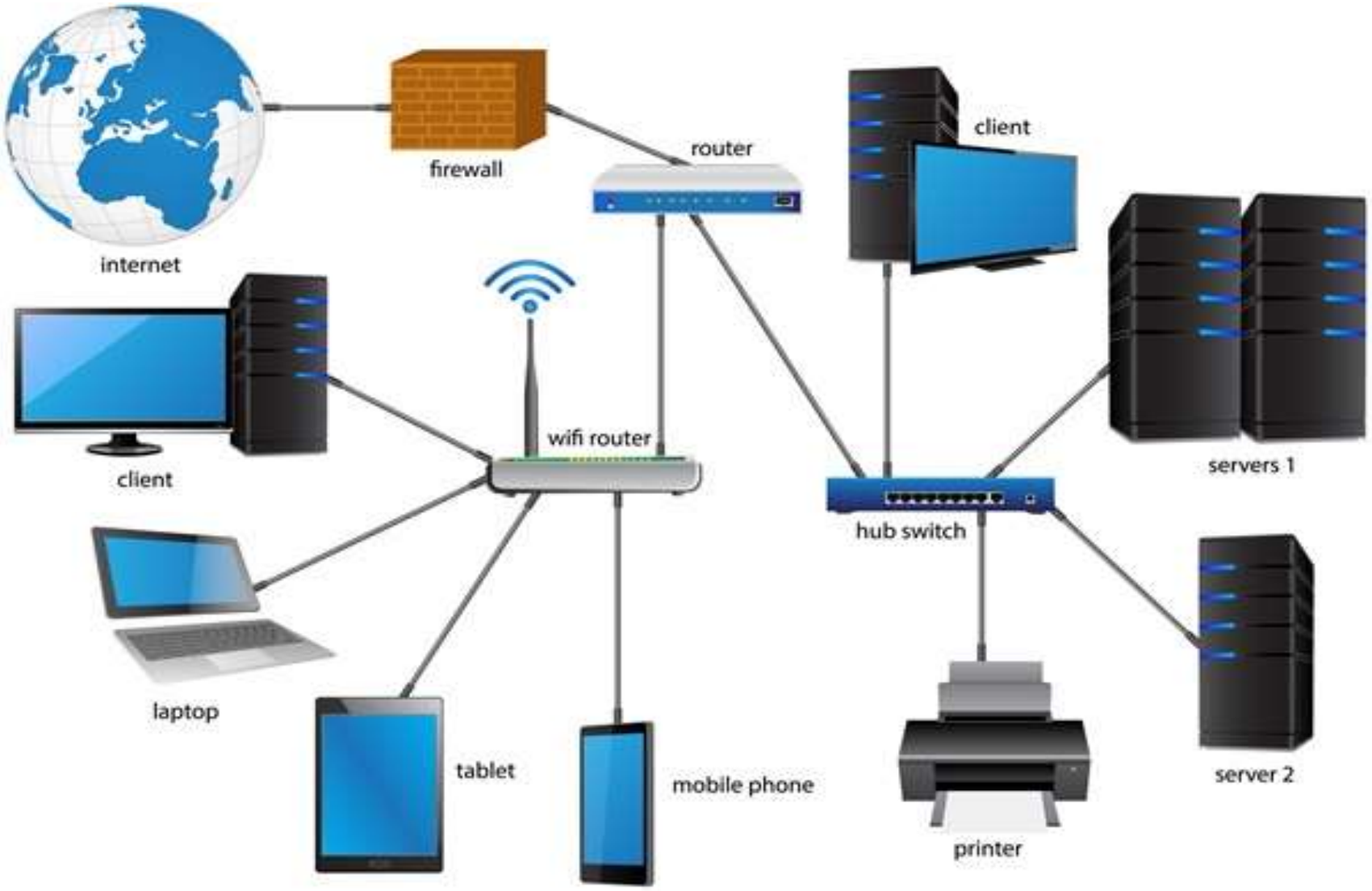
Wireless Home LAN



Wired LAN (Home Network) or HAN



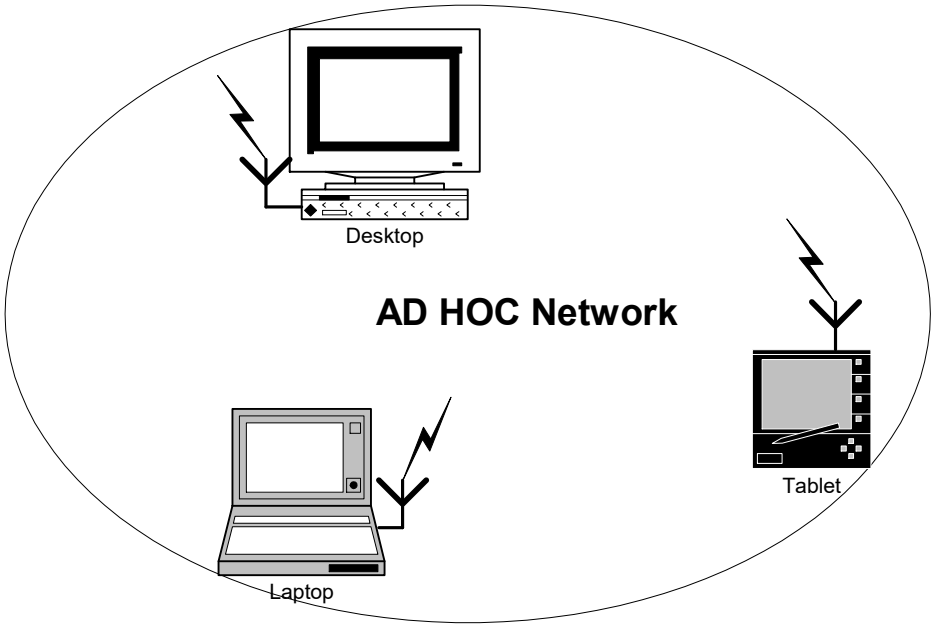
Wired and Wireless LANs having Access to the Internet



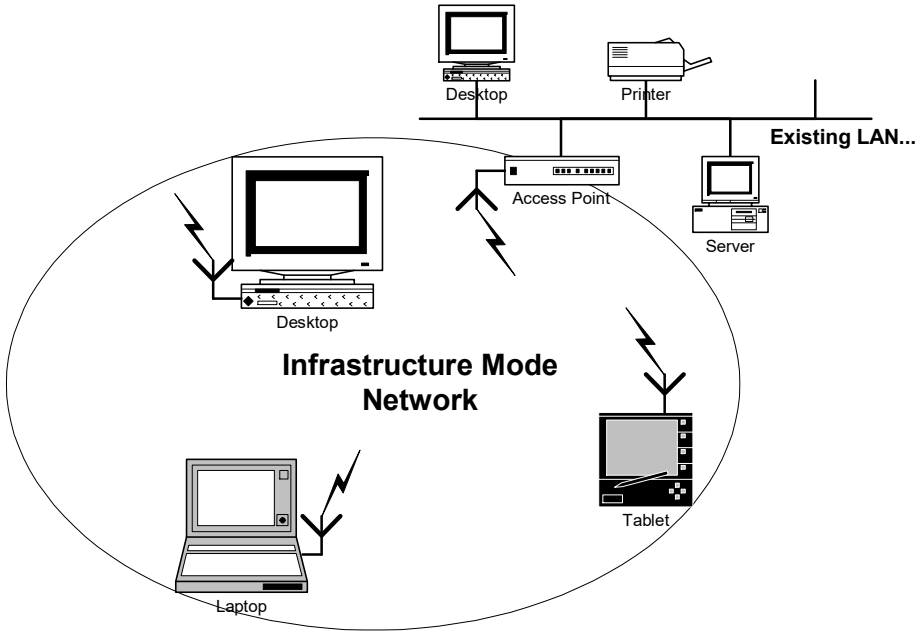
WIRELESS LANs - Wi-Fi

Provides a connection through an access point to the wider Internet.

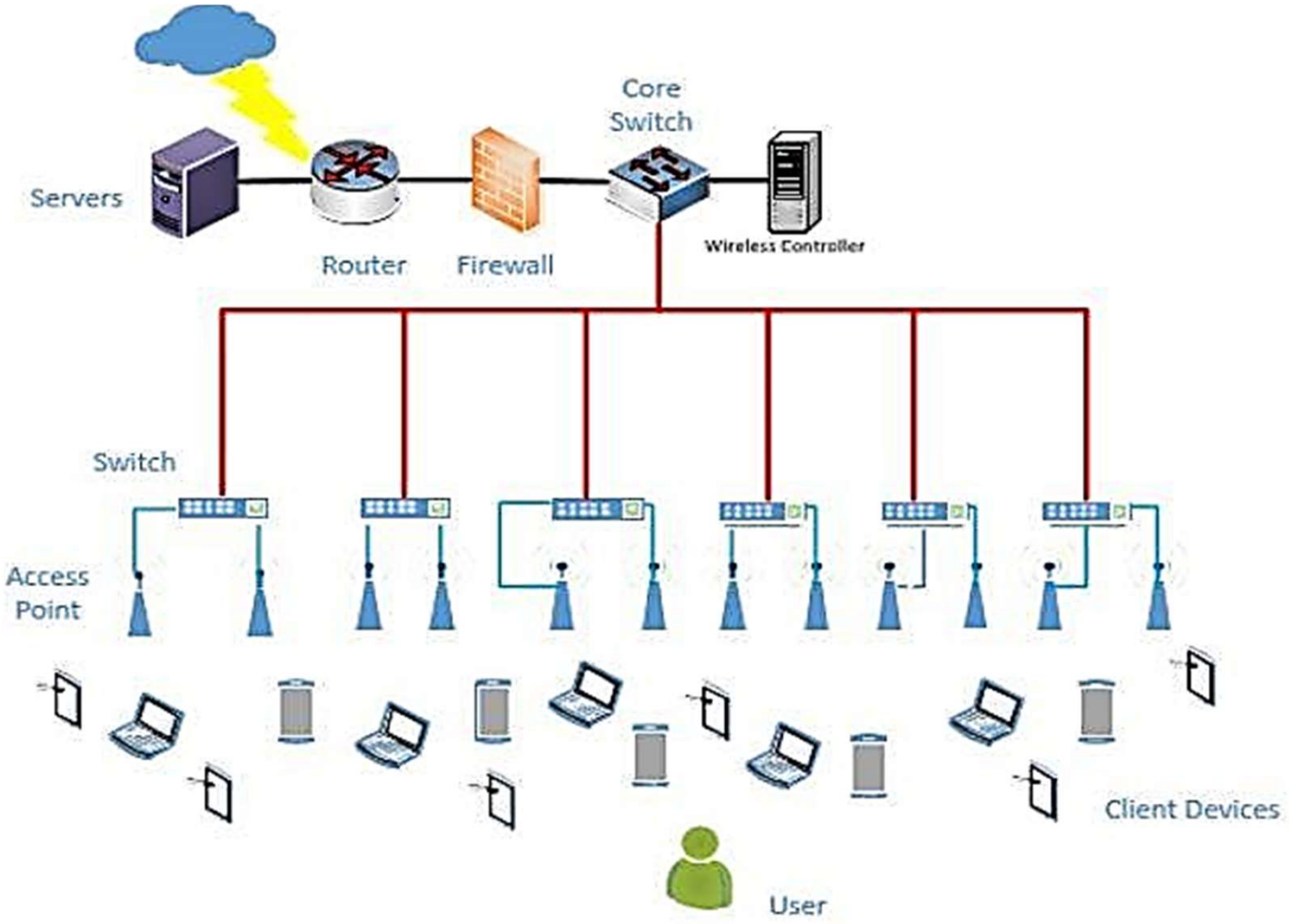
Independent Basic Service Set



Extended Basic Service Set

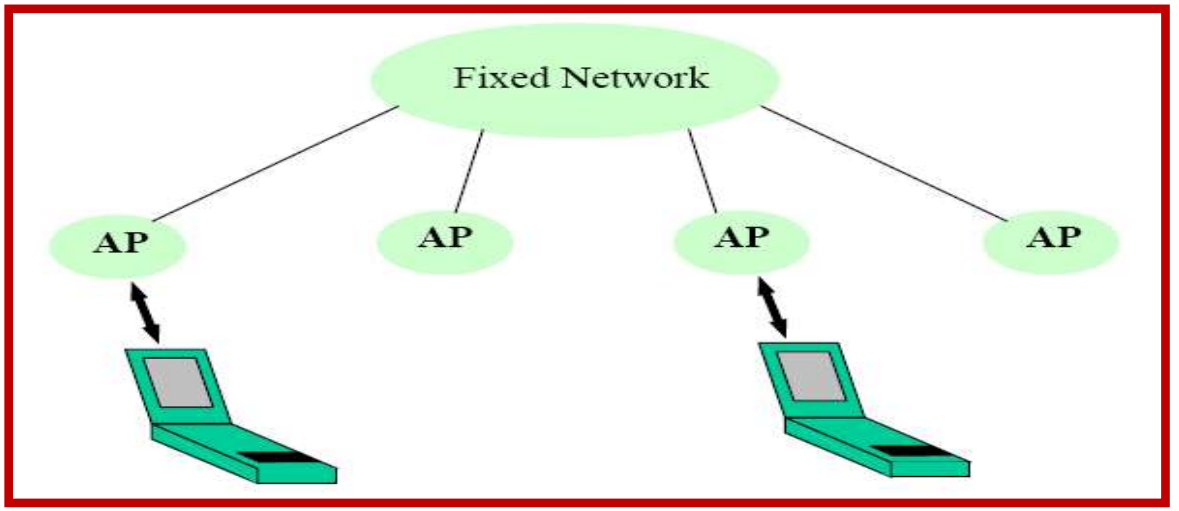


CAMPUS Wi-Fi NETWORK



HiperLAN 1/2

- High Performance Radio LAN (**HiperLAN 1**) standard
  - provide **high speed communications** (20Mbps)
  - **supports data such as video, voice, text, etc**.
- **HiperLAN 2**
  - provide high speed access (up to 54 Mbps) to a variety of networks including **3G mobile networks**, **ATM networks** and **IP based networks**, and also for
  - private use as a wireless LAN system.

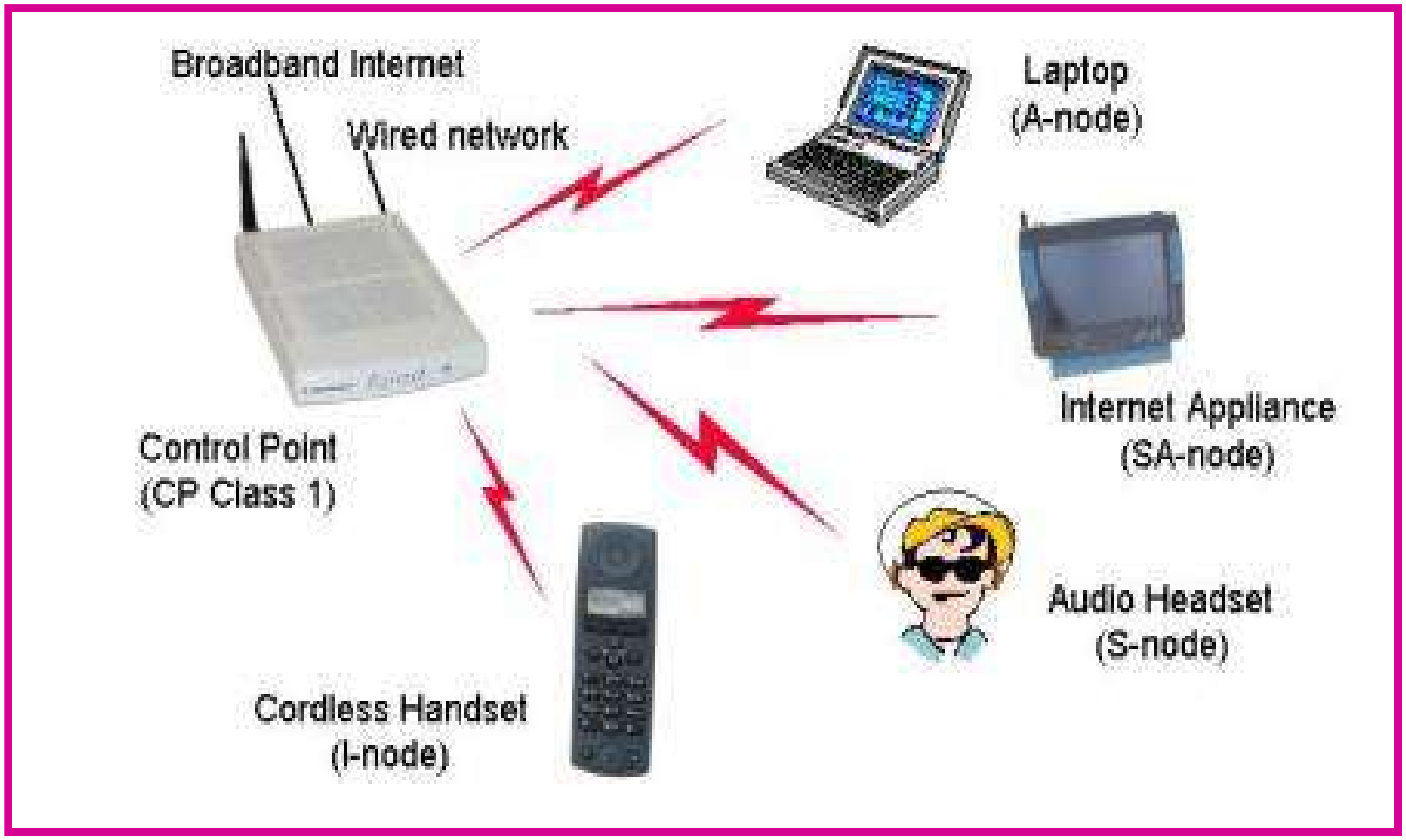


HomeRF

quality-of-service support for streaming media

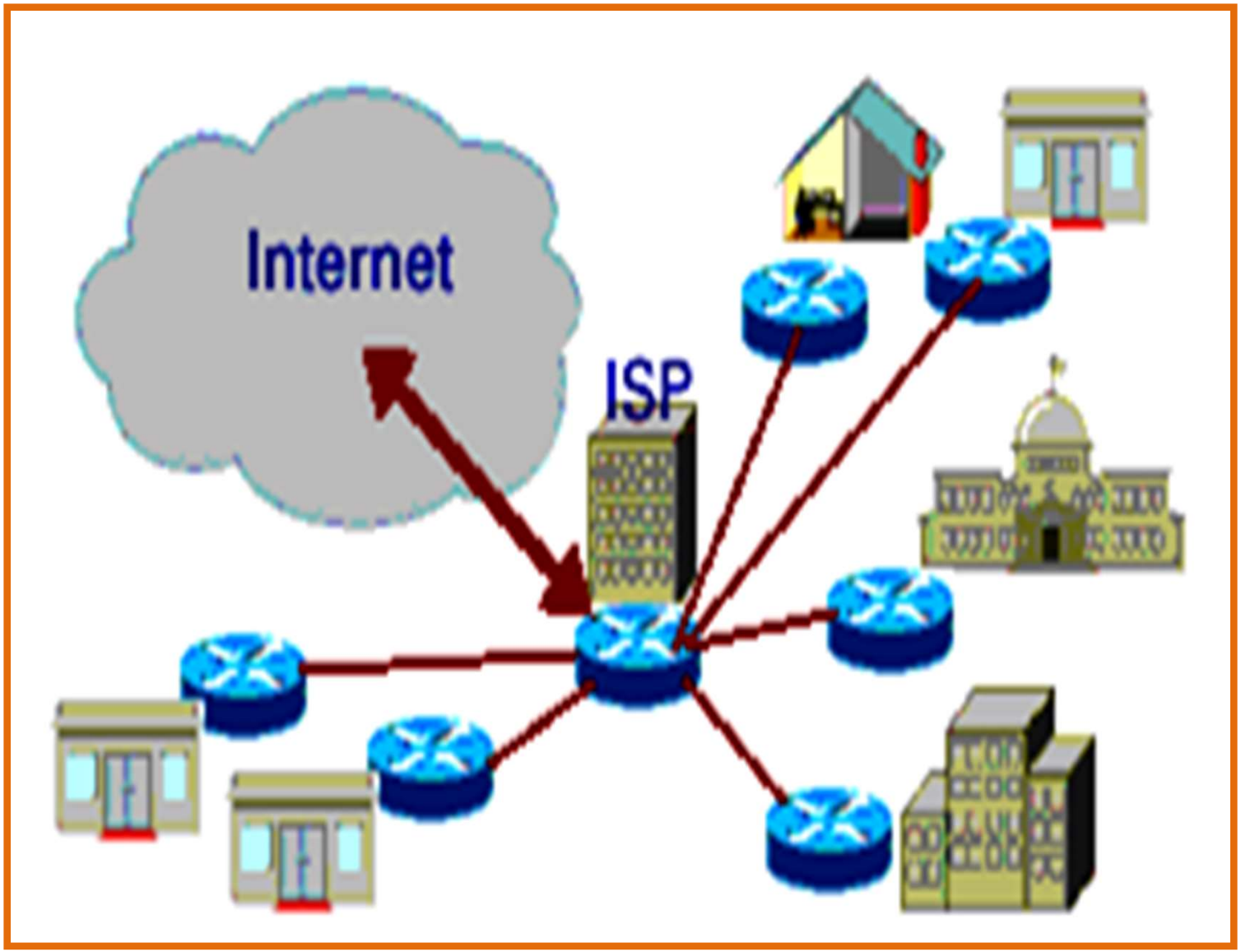
I – node (Isochronous) – Voice devices

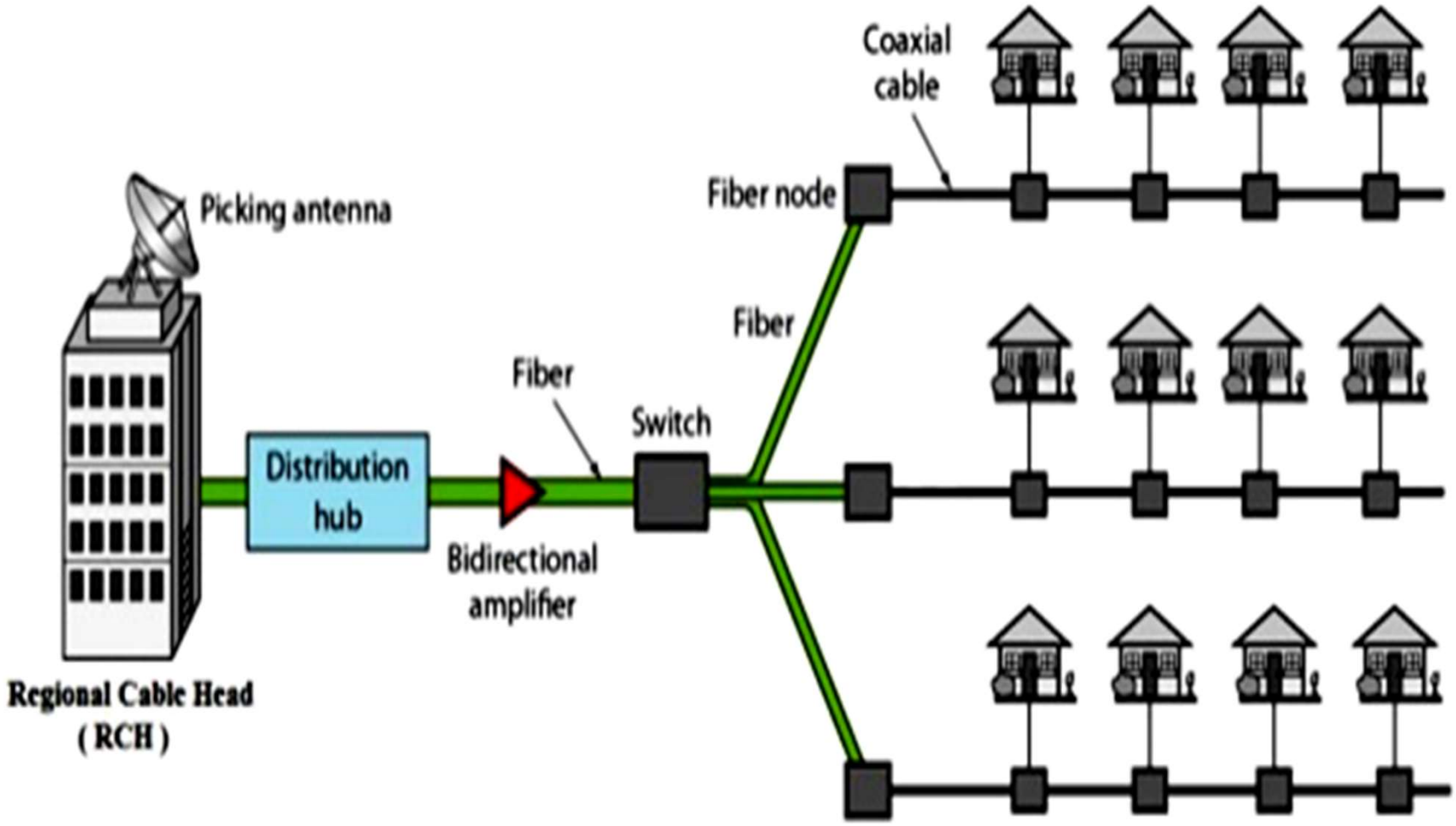
A node (asynchronous) - data devices



Metropolitan Area Network (MAN)

- **MAN** is a computer network larger than a Local Area Network, covering an area as large as an entire city, or a large campus
- When this type of network is specifically designed for a college campus, it is sometimes referred to as a campus area network, or CAN.
- A MAN usually interconnects a number of local area networks (LANs) using a high-capacity backbone technology, such as fibre-optical links, and provides up-link services to wide area networks (or WAN) and the Internet.
- A MAN may be wholly owned and operated by a private company, or it may be a service provided by a public company, such as a local telephone company.


Internet
ISP



Typical Cable TV Network

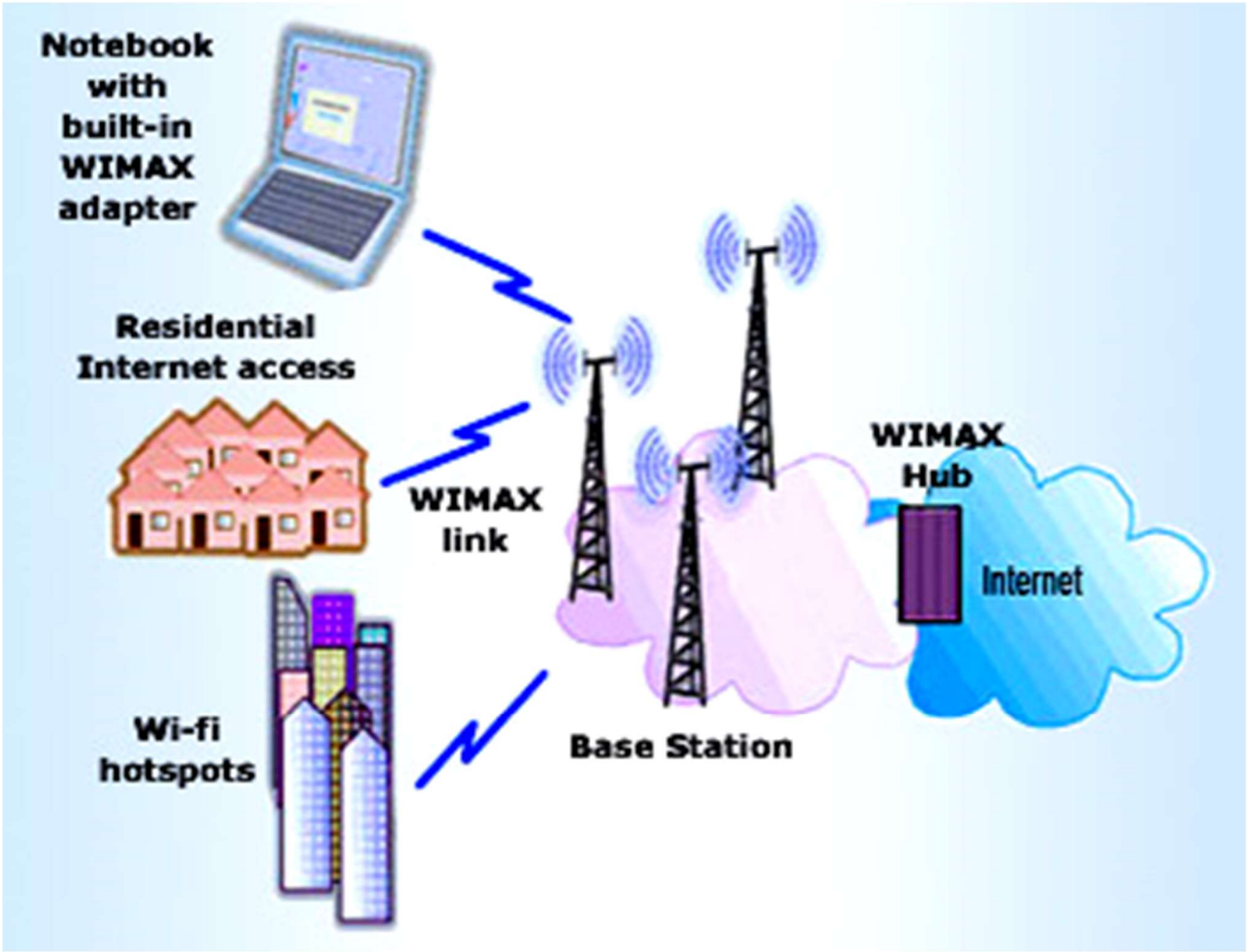
[An example of MAN]

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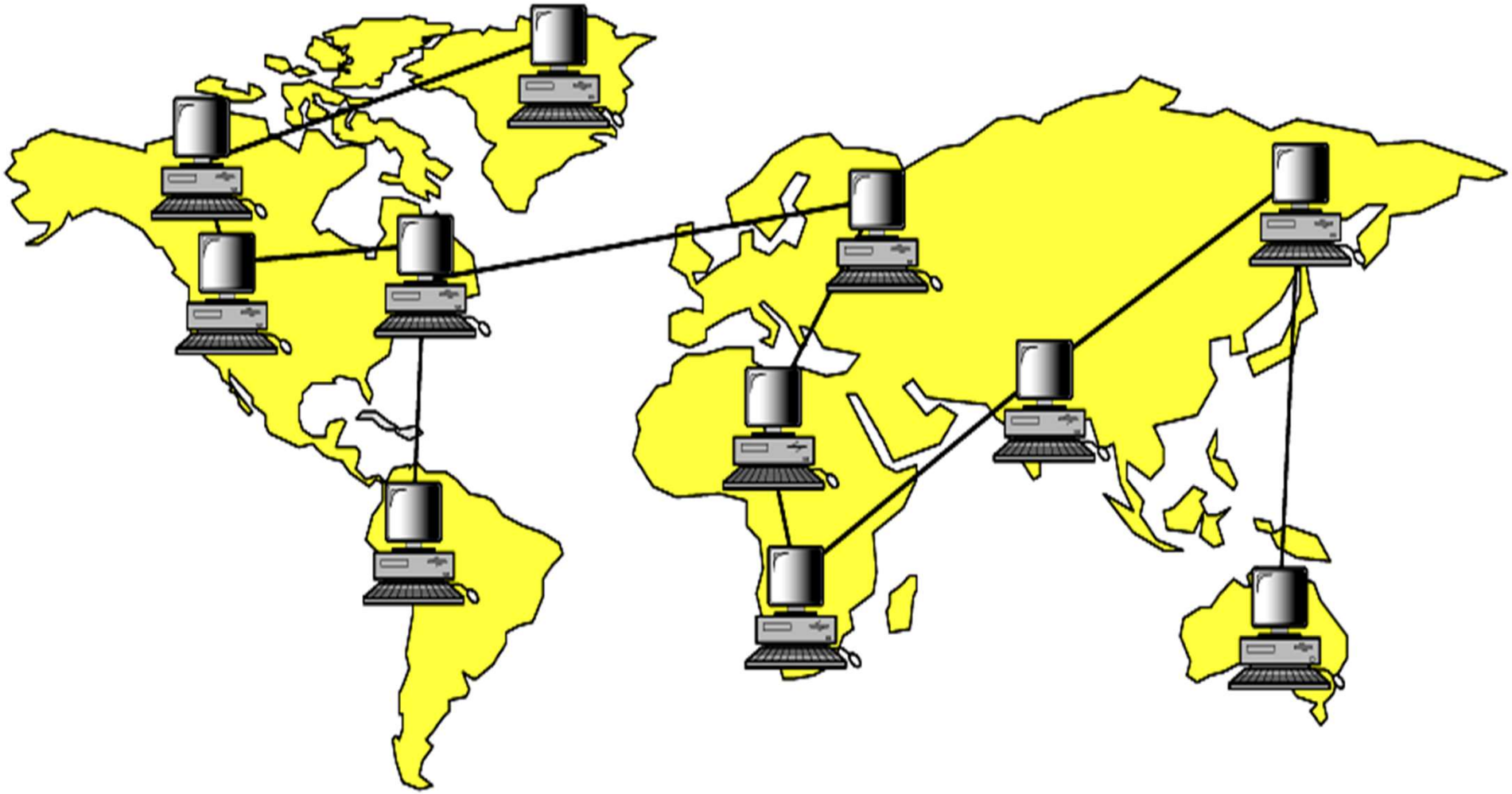
WiMAX - Worldwide Interoperability for Microwave Access

Metropolitan Area Network Technology

- Technology for wirelessly delivering high-speed Internet service to large geographical areas.
- "fourth generation," or 4G,
- Surpasses the 30-metre (100-foot) wireless range of a conventional Wi-Fi local area network (LAN), offering a metropolitan area network with a signal radius of about 50 km (30 miles).
- Enables the delivery of wireless broadband access as an alternative to cable and DSL".


Notebook with built-in WIMAX adapter
Residential Internet access
WIMAX link
WIMAX Hub
Internet
Wi-fi hotspots
Base Station

Wide Area Network



By Dr. Shuchita Upadhyaya Bhasin

Wide Area Network

- Networks that connect users over vast geographical distances are referred to as wide area networks (WAN).
- The communication links connecting nodes in a wide area network are either phone lines, microwave links, or satellites.
- For a computer to communicate with another computer over phone lines, a device called a **modem** is required. Modems can be internal or external.
- A modem (an abbreviation of modulator-demodulator) converts the computers digital signals into analog signals for transmission over phone lines (a process called modulation). The modem at the receiving end reconverts the analog signals back into digital signals that the receiving computer can understand.

Wide Area Network

- A Wide Area Network (WAN) provides long-distance transmission of data, voice, image, and video information over large geographical areas that may comprise a country, a continent, or even the whole world.
- A WAN can contain multiple smaller networks, such as LANs or MANs. The Internet is the best-known example of a public WAN.
- LANs are connected to WAN through a network device called router.
- WAN s use technologies like ATM, Frame Relay, X.25 etc.
- The communication medium in WAN can be PSTN or satellite links due to longer distances involved.
- WAN operates on switching.

WAN Topologies

- A **mesh or complete mesh topology** is also called a *point-to-point* topology.
- Each device is connected directly to all other devices on the network.
- **Mostly used in a wide area networks (WAN)**
- *Advantages* :

✓ the use of a dedicated links guarantees that each connection can carry its own load,

✓ Secondly, a mesh topology is **robust**. If one link becomes unusable, it does not incapacitate the entire system.

✓ Thirdly, when every message sent travels along a dedicated link, only the intended recipient sees it.. This ensures **privacy**.

✓ easy **fault identification and fault isolation due to** Point-to Point links.

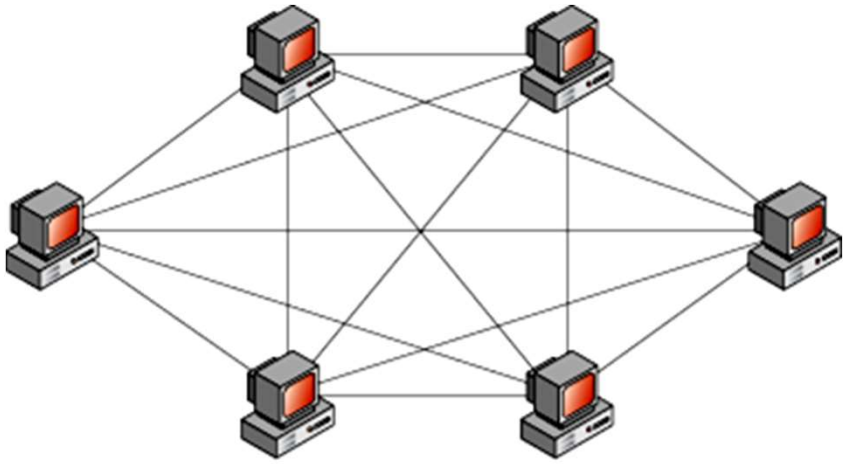
- ***Disadvantages :***

➢ **the hardware required can be very expensive** due to the requirement of big amount of cabling and a big number of I/O ports.

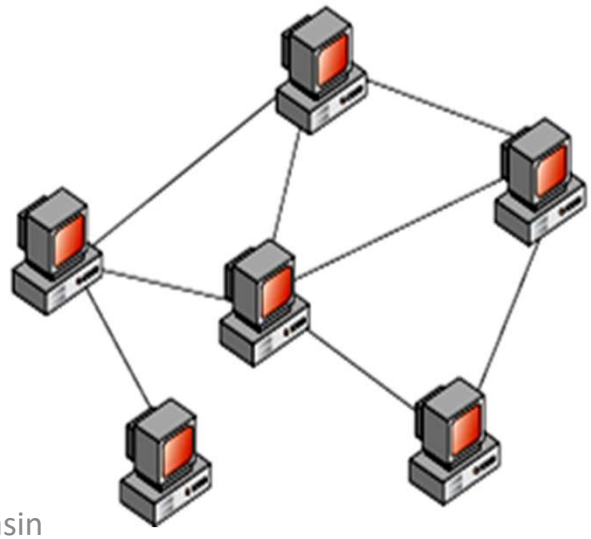
➢ **Installation and reconfiguration are difficult** because every device must be connected to every other device.

- For these reasons a mesh topology is usually implemented in a limited fashion called **partial mesh**
- instead of a one-to-one connection between every two devices, only a limited number of connections are made between the devices with the objective of attaining a connected network in which every node can be reached from every other node through direct or indirect links.

Complete Mesh

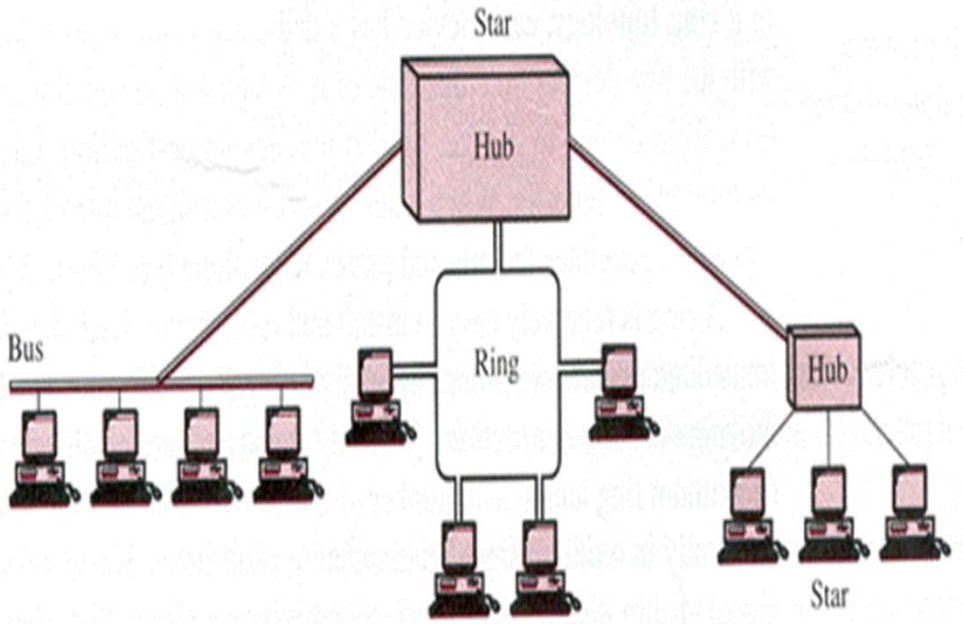


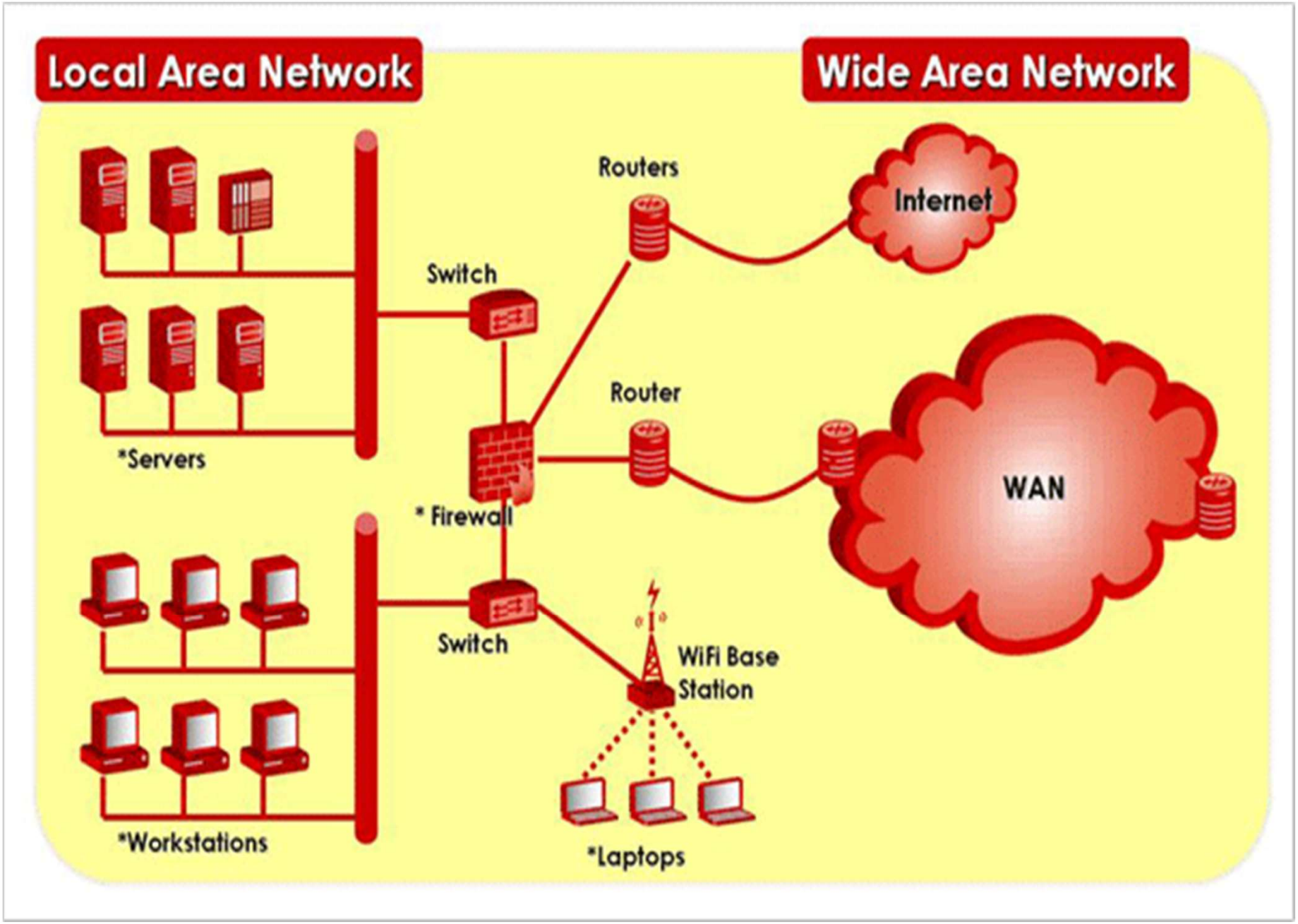
Partial Mesh



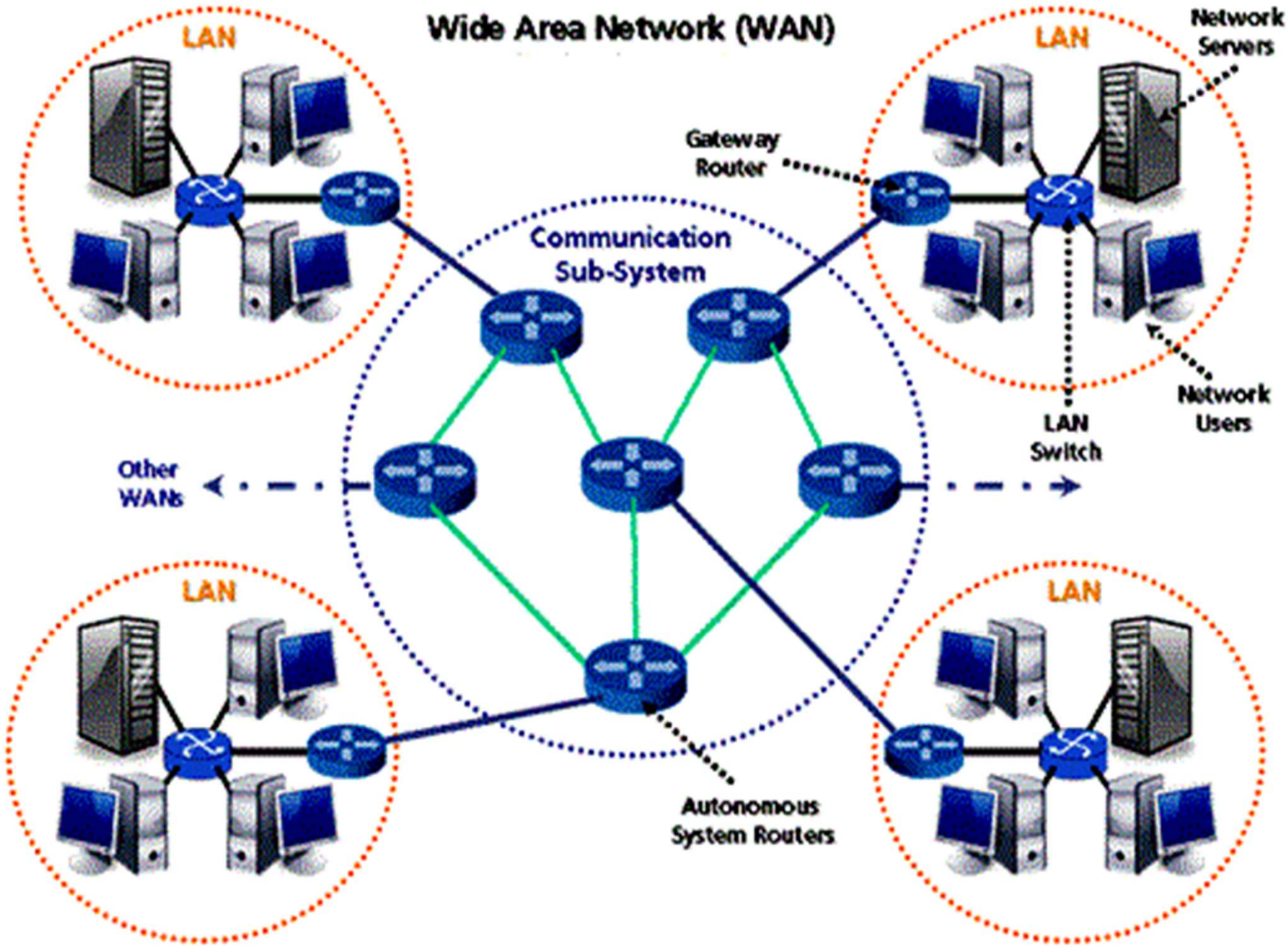
HYBRID Topology

- Not all topologies are strictly mesh, bus, star, or ring. Some variations of these topologies, known as hybrid topologies, are also used.
- These variations are combinations of two or more topologies and are called **Hybrid Topologies.**

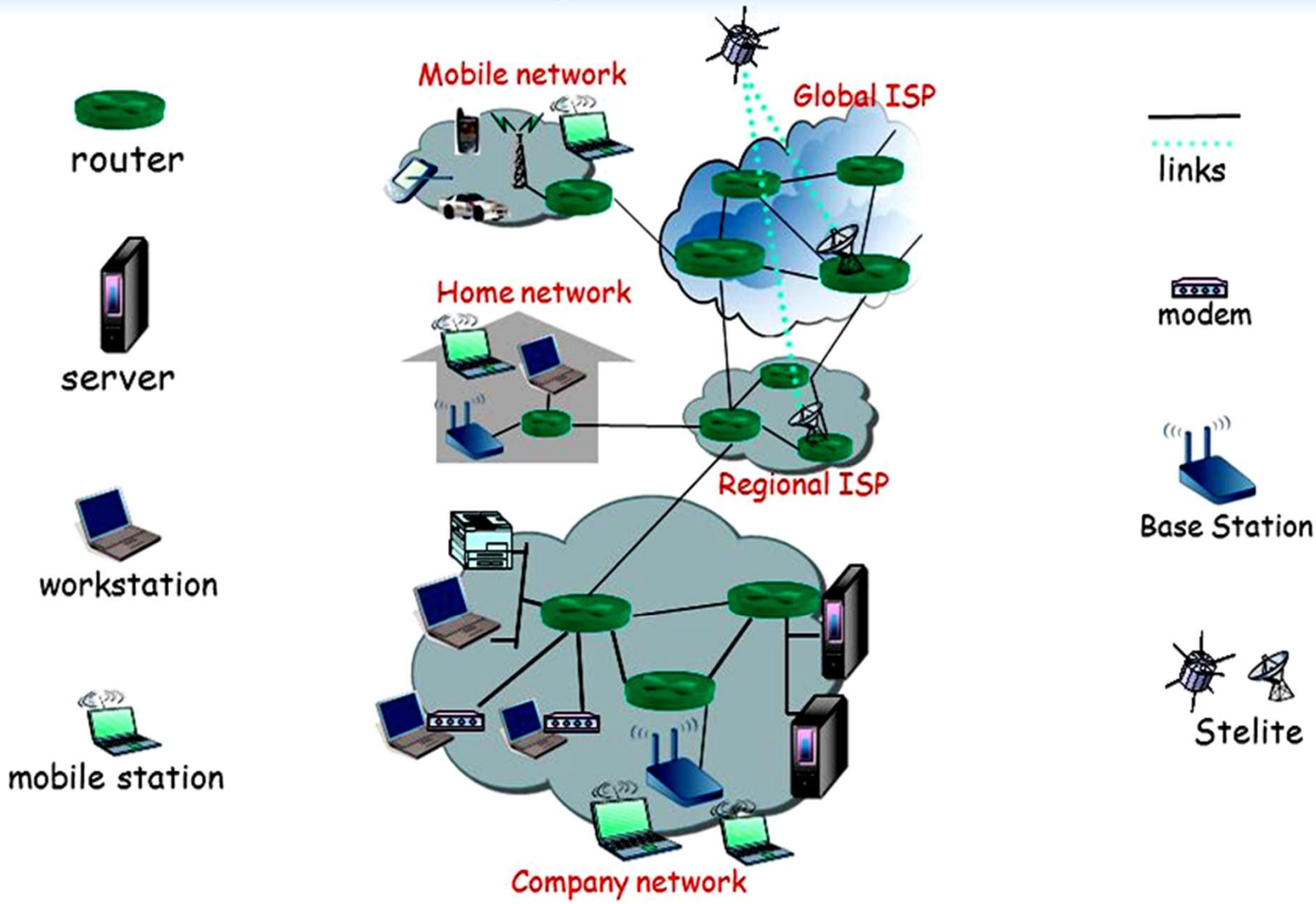




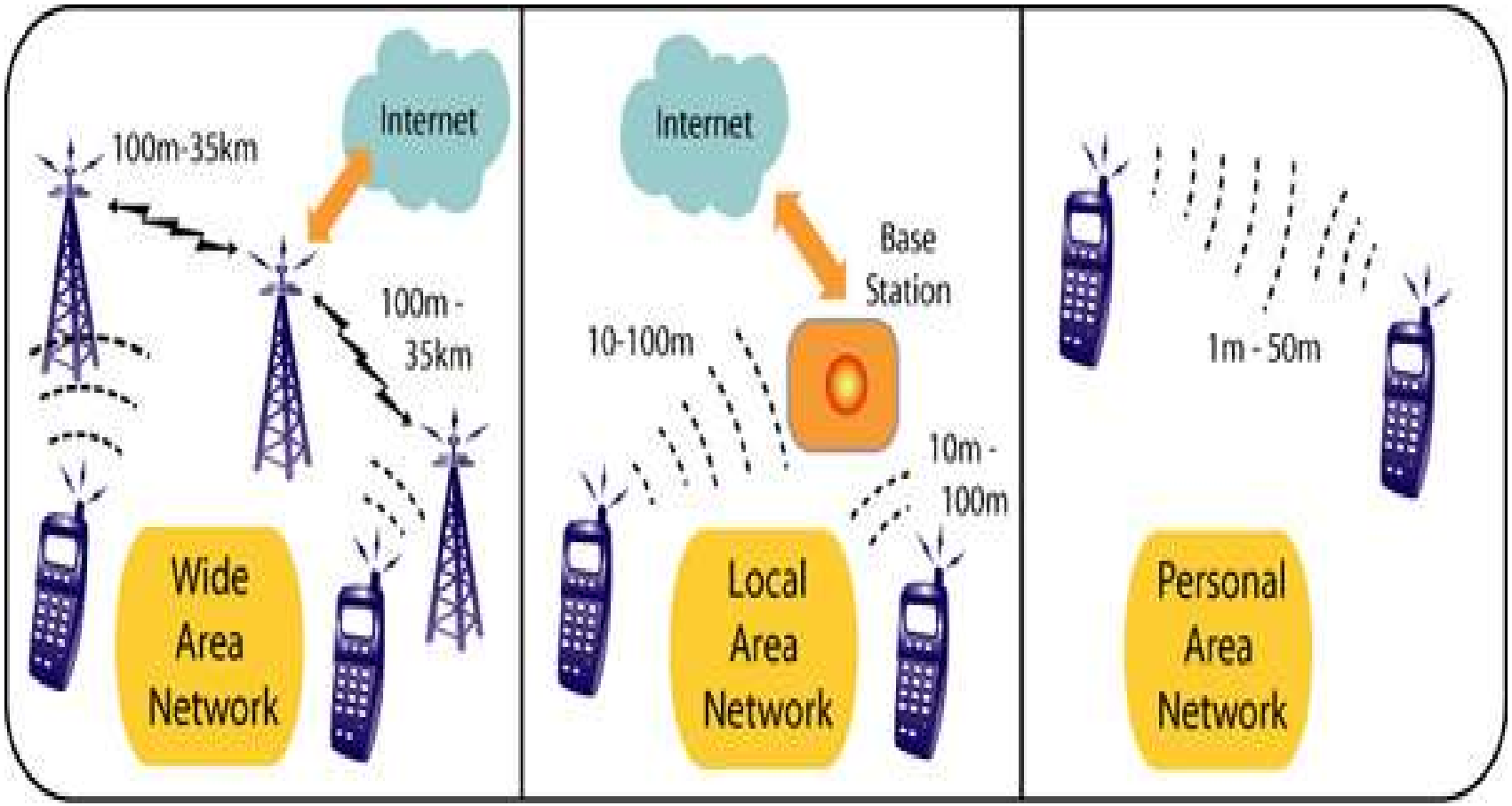
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Wide Area Network (WAN)
LAN
LAN
LAN
LAN
Network Servers
Gateway Router
Communication Sub-System
Network Users
LAN Switch
Other WANs
Autonomous System Routers

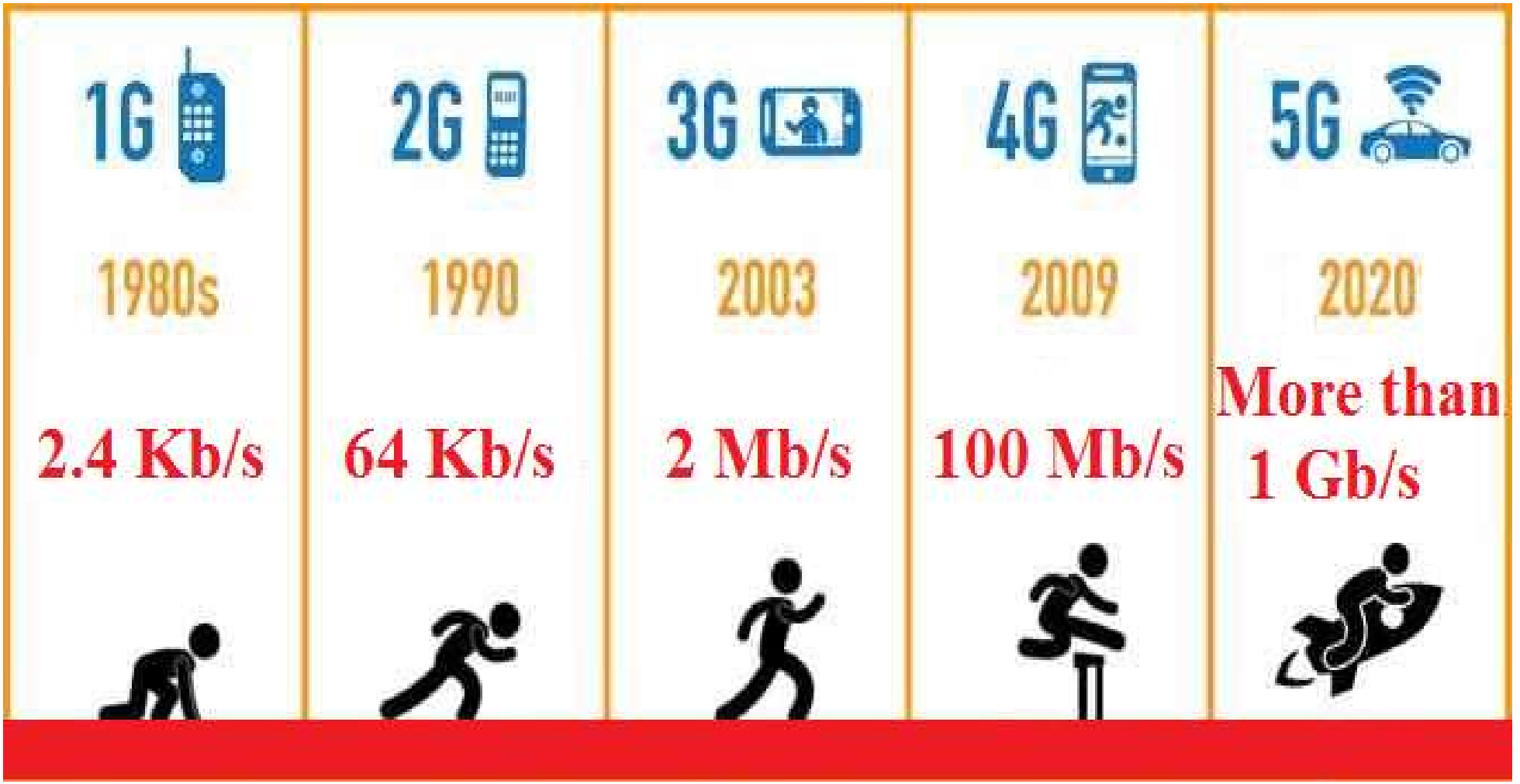
Internet: network of networks



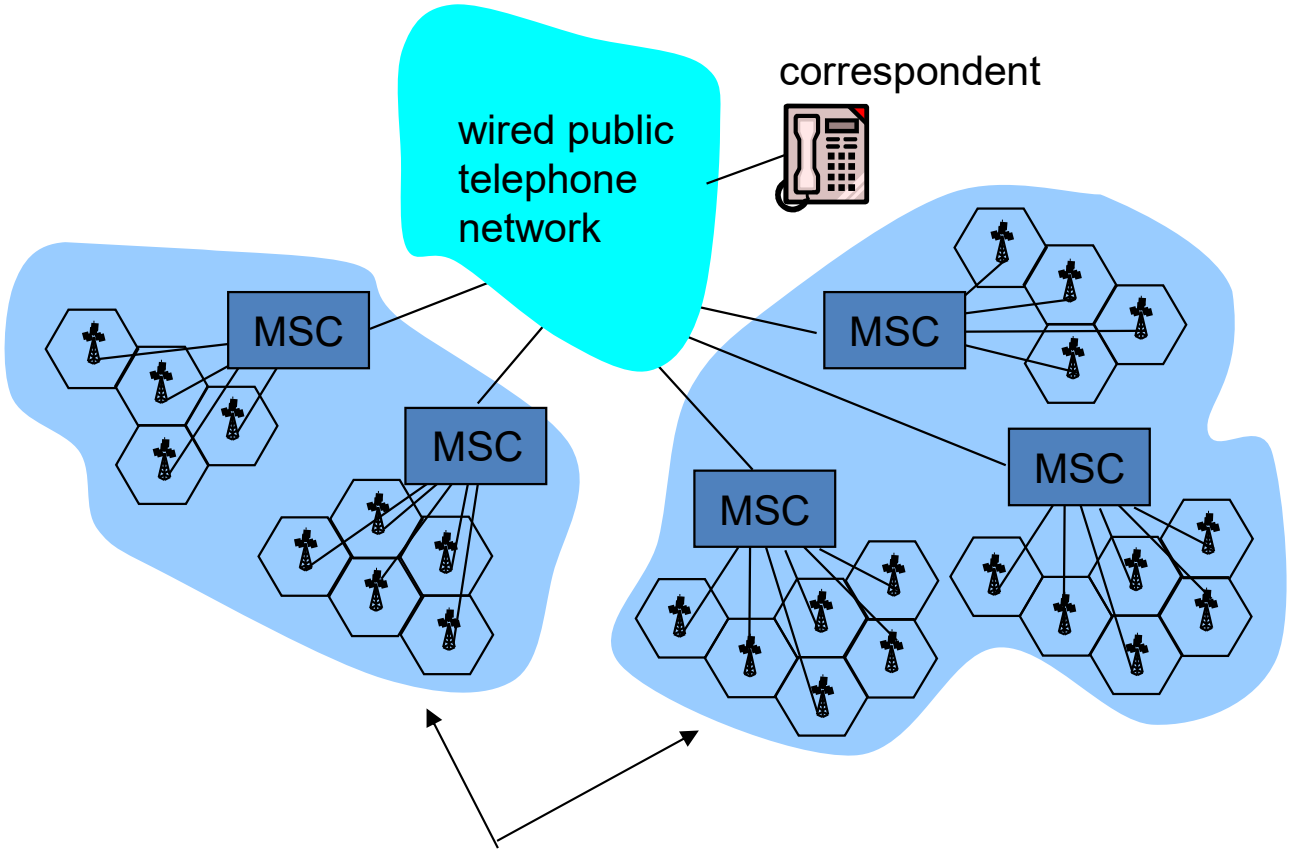
Wireless Networks



DIGITAL CELLULAR RADIO



cellular network architecture



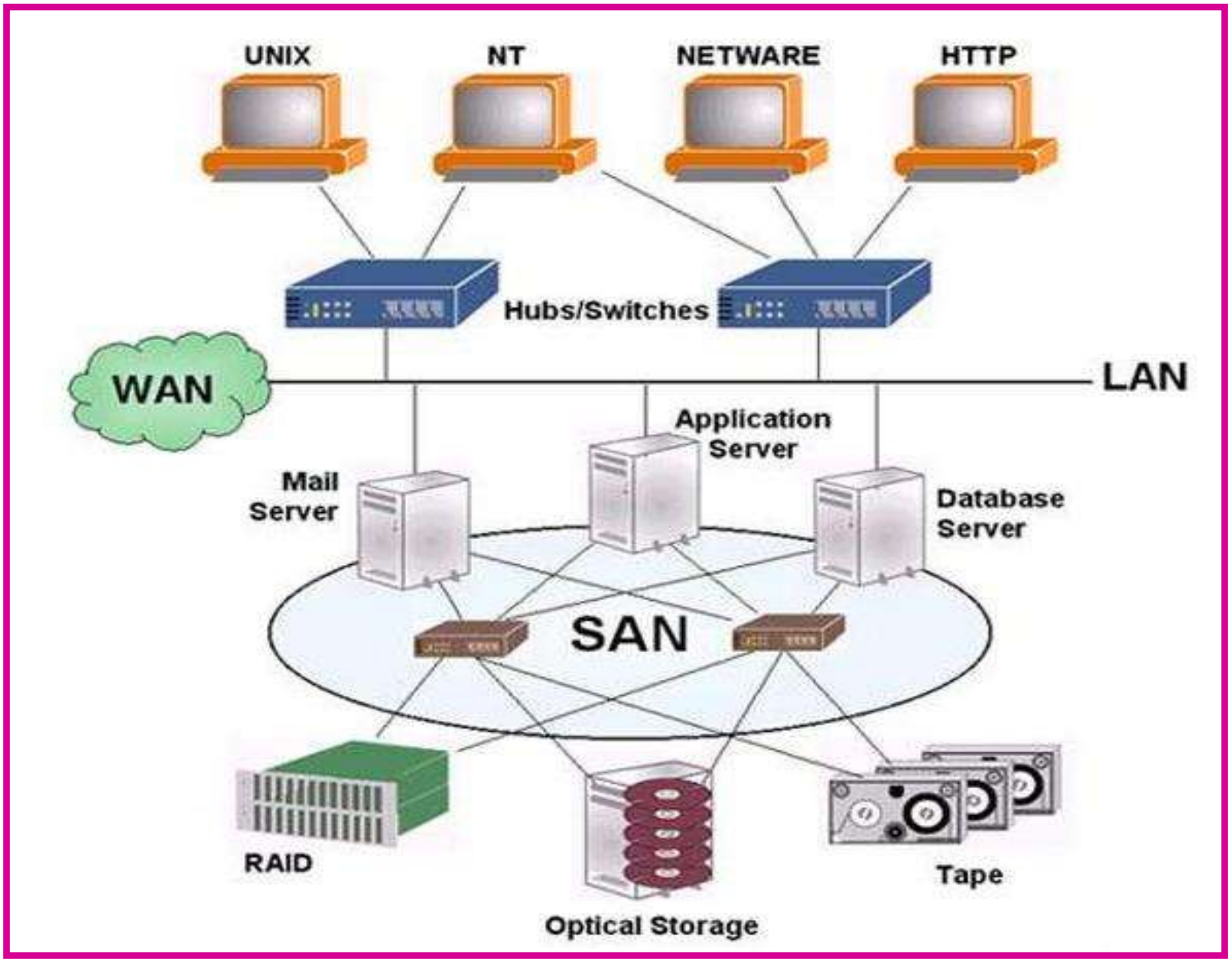
different cellular networks,
operated by different providers

Categorization Based on Purpose

Storage area network (SAN)
Enterprise private network (EPN)
Virtual private network(VPN)

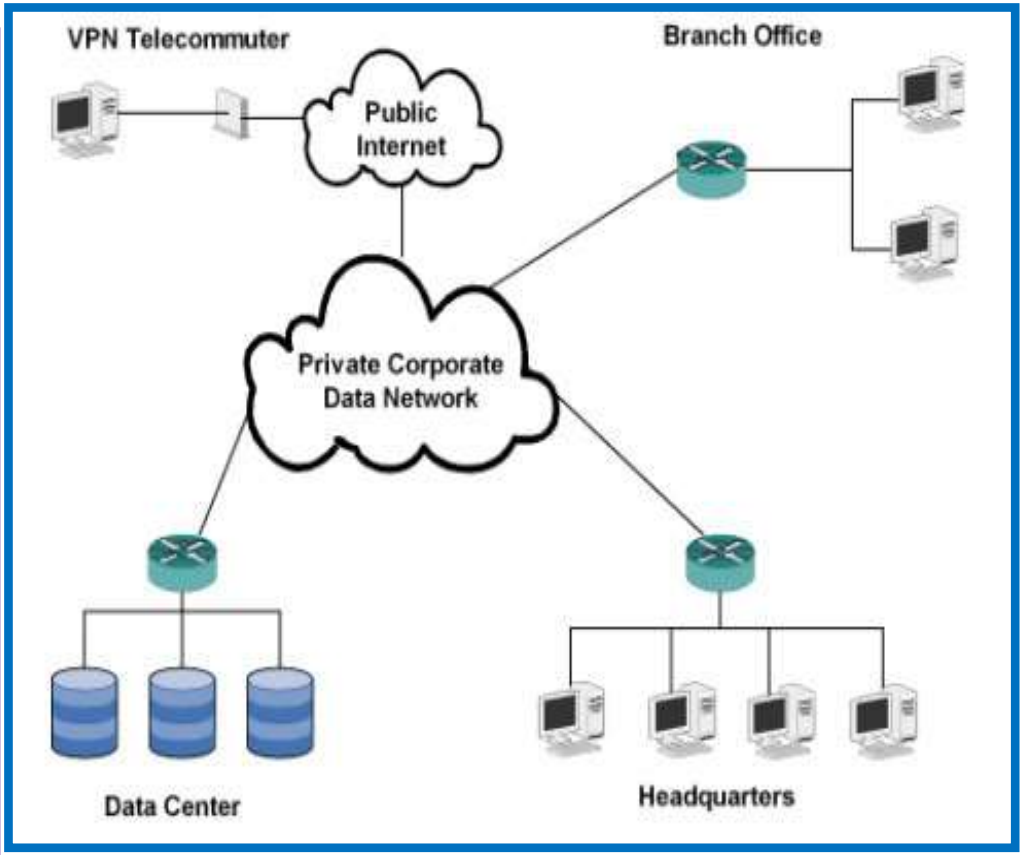
- Storage Area Network (SAN) is an **enterprise-level storage network** of several (often, hundreds) of storage devices (hard drives, disk arrays, tape libraries etc) managed by intelligent devices and accessible to servers so that the devices appear as locally attached to the operating system.
- Allows for maximally efficient use of all available storage space,
- **lowers total costs of ownership (TCO)**
- If a hard drive fails for some reason, the other hard drives in the storage network rebuild the data so no information is lost.

STORAGE AREA NETWORK



Enterprise Private Network (EPN)

- **Definition:** Enterprise private networks are secure, private communication networks established by businesses to meet their specific connectivity needs.
- **Ownership:** Owned and operated by the enterprise itself, providing control over network resources.
- **Scope:** Can cover a single office, multiple locations, or even extend globally, depending on the organization's requirements.
- **Security:** High emphasis on security, often using encryption, VPNs, and firewalls to protect data and communications.
- **Connectivity:** Provides reliable and dedicated connections for employees, partners, and customers.
- **Customization:** Tailored to the organization's needs, allowing customization of services, bandwidth, and infrastructure.
- **Applications:** Used for internal data sharing, VoIP, video conferencing, and access to enterprise applications.
- **Cost:** Requires significant initial investment and ongoing maintenance but offers greater control and security.
- **Examples:** MPLS, leased lines, and SD-WAN are common technologies used in enterprise private networks.

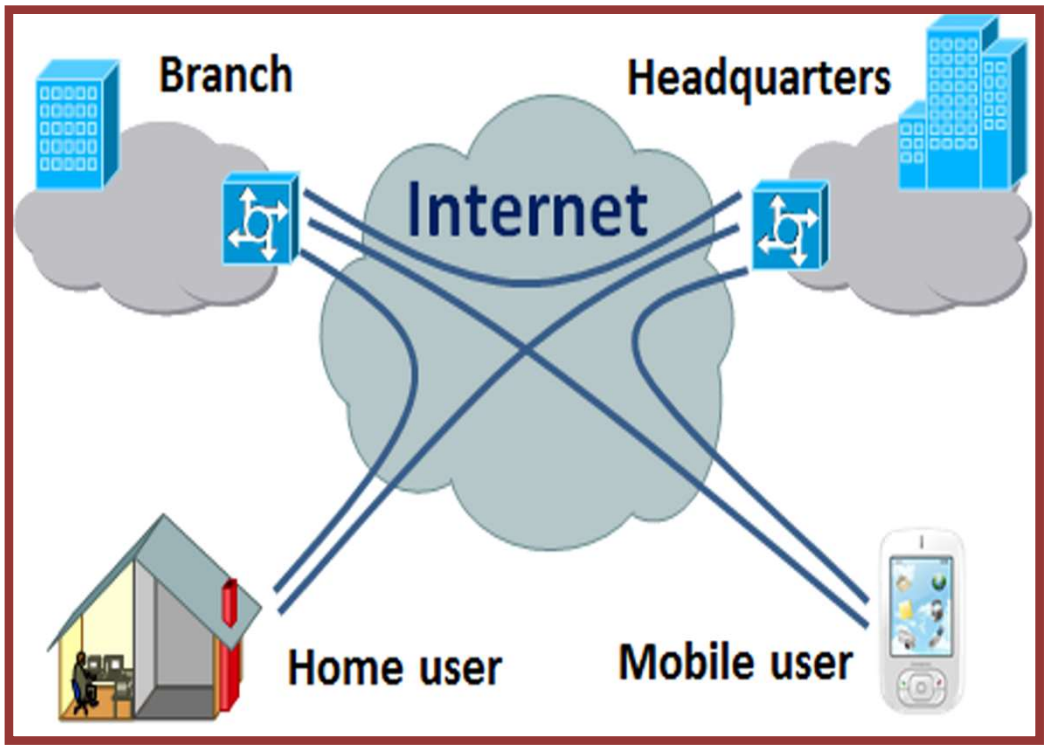


Enterprise Private Network (EPN) is a computer network that helps enterprise companies to connect their number of offices each in a secure way over a network.
Mainly set up to share computer resources.
Telecommuters can access the corporate network through a VPN that can be accessed via dial-up, DSL, cable modem, or other access method.

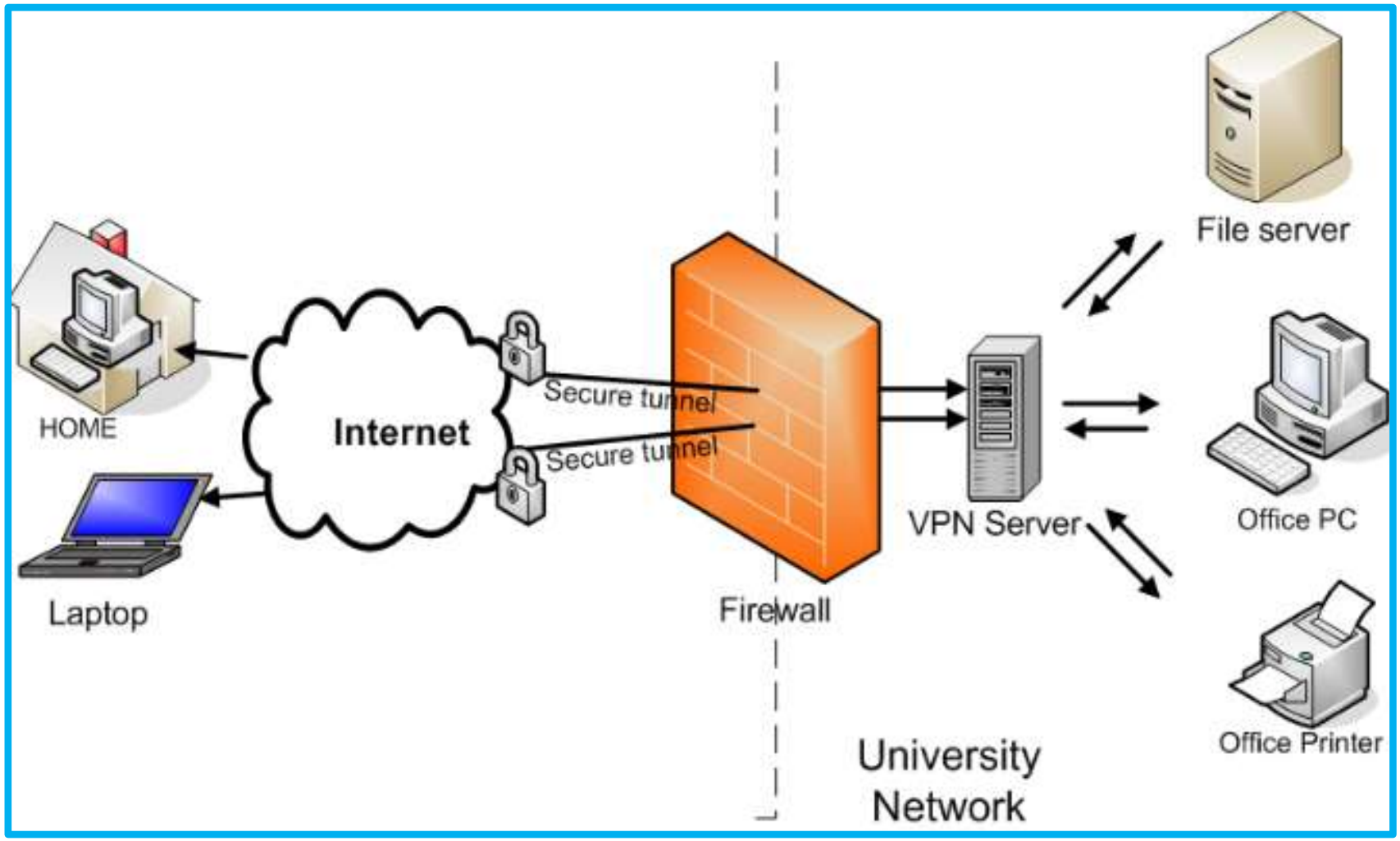
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Virtual Private Network (VPN)

- Virtual Private Network (VPN) is a network that is constructed by using public wires — usually the Internet — to connect to a private network, such as a company's internal network.
- It uses encryption and other security mechanisms to ensure that only authorized users can access the network and that the data cannot be intercepted.
- The first step to security is a firewall between the client and the host server - remote user establish an authenticated connection with the firewall.
- Encryption works by having all data sent from one computer encrypted in such a way that only the computer it is sending to can decrypt the data.
- A VPN might be used by teleworkers as a secure connection to their office. Students can use VPNs to connect to their school or university.



Virtual Private Network (VPN)



Distinction between LANs, WANs and MANs – detailed description is already covered in presentation 1

	LAN	MAN	WAN
Scale	Devices in a limited geographical area such as home, school, computer laboratory or office building	MAN is a computer network larger than a Local Area Network, covering an area as large as an entire city, or a large campus	A Wide Area Network (WAN) provides long-distance transmission of data, voice, image, and video information over large geographical areas that may comprise a country, a continent, or even the whole world.
Transmission technology	broadcasting	Usually broadcast	Point-to-point / store-and-forward/ Switched
Topologies	Bus, Ring, Star	Star, Tree	Mesh, Hybrid
Design issues	arbitration mechanism (Media Access Control) to resolve conflicts amongst multiple simultaneous transmissions over the single media, addressing, error control	Arbitration mechanism (media access control), addressing, error control	Routing, Congestion Control, Error control, flow control, addressing

SUMMARY

1. Data communication is the transfer of data from one device to another via some transmission medium.
2. A data communication network contains many different and varying components, the most common of which are the source, the transmitter, the transmission media, the receiver, and the destination.
3. Network architecture is based on the notion of layering. Layering involves combining network functions into groups that can be implemented together.
4. Each layer provides a set of services to the layer above it; each layer builds its services using the services of the layer below.
5. A protocol is a set of rules that governs how two communicating parties are to interact.
6. Different software design issues contribute to the successful transmission of data from the source to the destination in a network.
7. Interfacing is the connection of the communicating device with the transmission system and signal generation refers to the way in which bits (0 and 1) will be transmitted over the transmission system.
8. Addressing refers to the identity of the intended destination in a communication system. When a transmission facility is shared by more than two devices, a source system must address the intended destination so that only that system, receives the data.
9. Routing refers to the choice of an appropriate path from the source to the destination within a network with alternate connectivity.
10. Flow control is required to ensure that the source does not overwhelm the destination by sending data faster than it can be processed.
11. Error Control is required to overcome any errors that may occur in transmissions due to distortions of signals.
12. A network may provide reliability in terms of transmission such that even if an error occurs the system should be able to cope with it by either detecting and correcting it or providing ways for recovery.
13. Transmission System Utilization refers to how efficiently the transmission facilities may be shared among various communicating devices. Multiplexing and switching are two such techniques.
14. Media Access Control may be required in a broadcast network where the whole capacity of a single channel is shared amongst multiple contending users(computers) who are trying to access the channel simultaneously.
15. Security is required in a communication network to ensure that the data is received and used only by the intended recipient and is received by them in correct form.

SUGGESTED READINGS AND REFERENCES

1. Andrew S. Tanenbaum, “Computer Networks”, Pearson Education.
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5. Douglas E. Comer, “Internetworking with TCP/IP Volume – I; Principles, Protocols, and Architectures”, Fourth Edition, Pearson Education.
6. Relevant Websites: Online Resources and Web References.